MATRIX RING / AUTHOR READING EDITION

NEW RESEARCH BLUEPRINT

Generator Profiles of Galois Hulls of Principal Cyclic Codes over Matrix Rings

Recommended direction: P2-S — provisional
18 author decisions pending · 43 audit constraints retained
External novelty unverified · Phase 1 not authorized
Previous project remains on CONDITIONAL HOLD

Included in this PDF
Master overview and author-scope addendum first, followed by all nine detailed chapters and the complete audit-constraint crosswalk.

This is a read-only presentation of the existing planning files. It is not a research manuscript, proof, novelty certificate or permission to execute research.

From the current local Git workspace; **not committed or pushed**.
Branch: `arena/7d88376d-matrix-ring`
Original source files and sealed ZIP remain unchanged.

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Scope note. All ten original numbered documents are included in full. The later author-scope addendum remains an unapproved decision form. Historical words such as “current” or “now” are retained from their original documents; rendering this PDF does not change their authorization status.

Internal chapter links stay clickable. External URLs are retained. References to other evidence/scripts refer to the original sealed ZIP, not additional material reproduced in these pages. The source Markdown files and constraint CSV are attached to this PDF.

MASTER NEW RESEARCH BLUEPRINT

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Generator Profiles of Galois Hulls of Principal Cyclic Codes over Matrix Rings

Working title — Arena recommendation only. Author selection and all research phases are pending.

> **BLUEPRINT DEVELOPMENT ONLY. Wait for explicit author approval before beginning Phase1.**

Executive decision brief

Recommend **P2-S: a structural, principal-input Galois-hull programme**, beginning at matrix rank2 with a bounded nilpotent/repeated-root local model. The intended new paper would determine **when a principal input retains a principal hull, or which structural mechanism permits generator growth**, with an explanatory criterion/classification. It would not repeat MS03's matrix/Galois/Gray/quantum chain, repair it under a new title, or treat larger matrices as the main contribution.

Candidate	Advisory score /100	Disposition
P1 — non-reflexive physical/quantum mechanism	50.8	Reserve alternative; substantial internal/external overlap
P2-S — structural principal-input hull profiles	**73.6**	**Recommended for a literature-first gate, NOT selected or approved**
P3 — skew/Galois matrix compatibility	65.6	Reserve alternative; new compatibility theory required, not ingredient novelty

These are subjective weighted planning scores, not measured novelty or acceptance probabilities. The exact external gap remains **unverified**. A strong competing theorem or a routine-reduction result would change the recommendation.

Read the blueprint

1. Candidate directions and weighted scoring
2. Literature and novelty positioning
3. Recommended problem — selection pending
4. Mathematical framework and inherited inputs
5. Prospective theorem roadmap
6. Computational methodology and prospective tables
7. Optional quantum strategy
8. Complete phase roadmap and measurable exits
9. 18 pending author decisions
10. This master document.

1. Title

Generator Profiles of Galois Hulls of Principal Cyclic Codes over Matrix Rings. The title is neutral about whether growth occurs. Stronger preservation/growth or quantum titles require actual results and later author approval.

2. Motivation

MS03 and the audit already establish a framework in which cyclic right ideals correspond to embedded modules and hulls are intersections with a specified dual module. In a non-chain local factor, a submodule of a two-generated module can require more generators; the special relationship to a dual may impose additional restrictions. Explaining that restriction or growth mechanism is the proposed mathematical motivation, not merely constructing more codes.

3. Verified literature baseline

Published anti-isomorphism/Frobenius duality, field/commutative-ring Galois hulls, repeated-root scalar hulls, MT/QC hull generator matrices, Artinian μ/type/dual invariants and skew/Galois combinations are occupied terrain. The bounded fresh review adds or sharpens NL01–NL11; the inherited source/cell ledgers and A8 matrix-ring Gray/CSS predecessor remain binding. 02 separates primary-record/preview/selected-text evidence from complete proof applicability.

No “first matrix-ring quantum”, “first q8 EA”, generic hull-generator, or ingredient-only skew×Galois claim is available. Several primary technical comparisons remain necessary. A11 metadata, A5a author order and the MT2025/2026 version relationship remain qualified; missing metadata is not evidence of nonexistence.

4. Precise proposed gap

Internal unresolved question: can a principal cyclic code have a nonprincipal Galois hull in the repeated-root nilpotent setting (MS03 O-3.1b)? Known nonprincipal hull witnesses start from nonprincipal codes. The inherited q4,t2,n2 census does not settle the question over other fields or multiplicities.

Proposed differentiated gap: a structural generator-profile/principality criterion for that principal-input subclass, retaining the embedding and induced-form data that coarse dimensions forget.

External status: X — NOT CERTIFIED UNOCCUPIED. Scalar classifications, GPM/intersection methods or Artinian algebra may already imply part or all of it. Phase1 must test those reductions before proof/implementation effort. Internal openness does not establish external priority.

5. Primary objective

Seek a nonroutine theorem/classification explaining the attainable generator profiles of $H_s(W)$ for $\mu_B(W)\leq 2$, and hence principal-hull preservation versus generator growth. Initial proposed class: $q=2^m, m\geq 2$; $r=2, t=2, n=2$; all s. Any extension to $e=p^a>2$, multiple factors or $r\geq 3$ is conditional, not an automatic parameter enlargement.

6. Research questions and conjecture posture

Core questions RQ1–RQ8 in 03 ask: existence/preservation, sufficient embedding/profile data, sharp spectrum, twist dependence, global coupling and structural certificates. RQ9 is optional physical relevance.

PC1 is an explicitly exploratory principal-hull-preservation conjecture in the smallest proposed local family. It is neither adopted nor proved; only the old q4 evidence supports that instance. Arena recommends question-only posture until the gap check. A counterexample, a preservation theorem, an

already-known result or a no-go outcome are all possible. No later result may assume PC1 or old H-3A.1/H-3A.2.

7. Algebra and standing assumptions

q=p^m; A_t=F_q[u]/(u^t); R=M_r(A_t), primarily r=2; X central; B=A_t[X]/(X^n−1); S≅M_r(B); **right ideals**; first-slot θ_s=σ_s∘transpose with u,X fixed; rev is coefficient reversal. Field degree m and matrix size r are not interchangeable.

Known positive DD is +σ_(2s)(C), not unconditional reflexivity. Product-of-chain-ring generation is not freeness. The side, slot, trace level and exact induced form are part of every statement. These are inherited inputs, not the proposed new contribution.

8. Definitions and framework

For the column Morita module W, I_W={M:im M⊆W}; principal iff μ_B(W)≤r. Define D_s(W) as the module of the **actual global dual**, H_s(W)=W∩D_s(W), and record generator growth Δ_s(W) and principality margin μ_B(H_s(W))−r. Local factors use a source-checked model K[u,w]/(u^t,w^e) and induced maps/forms, not an invented standard dot product.

These definitions are research bookkeeping, not claimed novel invariants. A normal form must transport the pairing and retain embedding information; arbitrary ambient GL operations are not automatically isometries. Details and non-waivable exclusions are in 04.

9. Theorem roadmap

Targets	Role	Current status
NT01–NT02	Valid inherited inputs and faithful local/form specification	Pending infrastructure; normally not novelty
NT03–NT04	**Generator-profile structure and principal-input criterion/dichotomy**	**Core proposed research; truth and originality unresolved**
NT05	Sharp bounds/spectrum or preservation boundary	Conditional proposed result
NT06–NT07	Twist/embedding and nonroutine global synthesis	Optional justified extensions
NT08	Correct certificate-producing structural method	Conditional; not implemented
NT09	New physical consequence only through a valid bridge	Optional; no bridge established

Every future theorem needs an exact statement, complete proof, source-reviewed dependencies and separate novelty comparison. Finite tests cannot promote a target/conjecture to theorem. 05 specifies obligations and failure outcomes without providing proofs.

10. Expected contributions

The intended contribution is **explanation and classification**, not a parameter scan. A paper-sized outcome would contain at least one nonroutine structural main theorem plus a meaningful proved consequence/classification, with explicit sharpness/preservation examples and honest predecessor differentiation. Routine Nakayama/Morita/CRT facts, an isolated witness or old-paper proof repairs are insufficient alone. Success is not promised.

11. Computational methodology

Design only: separately specified field/coordinate contracts; definition-based and module-structural hull paths compared as **subspaces**; residue/radical minimal-generator certificates; independent arithmetic; explicit family/sample/timeout labels; negative controls for sides, reversal, forms and representation lengths.

Proposed future pilot cap: ≤3 families, ≤2,000 new distinct inputs/family, total2 CPU-hours/4GiB/250MiB, subject to explicit approval and preflight. No scientific scripts are imported or executed now. Old finite campaigns are reused as evidence rather than gratuitously rerun. 06 contains the method, not an implementation.

12. Quantum strategy, only if justified

Default: algebra-first; quantum optional. Coefficient expansion transports to the specified J/s form, not automatically Euclidean. Standard Euclidean EA uses **h_E**, c=k−h_E, K=N−k−h_E and a valid Bob completion, not μ_B(H_s). SO/DC/R1 have different stabilizer inputs/orthogonals/logical-error sets. General trace/additive labels require Fp-level treatment.

No distance, record, low-entanglement ceiling or “first q8” claim is made. If no new metric-qualified physical consequence survives QG1–QG6, omit the quantum track. 07 supplies its separate gates.

13. Expected examples and parameter tables

Planned tables record actual input/hull module bases, μ, dimensions, local layers/relations, embedding/form data, proof locators and certificate status. Boundary, LCD/SO, preservation/growth and coupled-factor examples are conditional on existence and results; no new row is populated with invented values. Any quantum table additionally separates physical alphabet, field-code alphabet, route,K,c and exact d versus a lower bound.

The old28 exact supplied matrices,51 selected checks (48 exact/3 lower-bound distances),1,221-module census and742 DC bookkeeping failures remain **inherited audit evidence**, not new-paper discoveries or counts of false theorems.

14. Novelty comparison

P2-S must differ from MS03's arbitrary-ideal framework and nonprincipal-input examples, NL10's scalar repeated-root hulls, NL03/NL04's field hull/intersection generators, and NL11's Artinian invariants/duality. If the proposed criterion is a short standard specialization, it is not a new main contribution. P3 cannot be justified by combining skew and Galois words; NL05/NL06 already constrain that position. 02 prescribes exact theorem-by-theorem comparison, not a failed phrase-search argument.

15. Risks and alternatives

Main risks: externally occupied or routine question; full-wild classification complexity; insufficient coarse invariants; incorrect local/form transport; no useful quantum bridge; uncertain old source/editorial facts. Alternatives are explicit restriction, a meaningful preservation theorem, a structural growth theorem, algebra-only scope, re-scope after comparison or stop. Increasing matrix size or hiding audit defects is not a fallback.

16. Q1-journal positioning

Provisional first **scope** target: Designs, Codes and Cryptography for substantial algebraic coding structure. Conditional alternative: Finite Fields and Their Applications if Frobenius/factor/finite-field content is genuinely nontrivial. Official scope pages were checked at recorded levels. **Current Q1 quartiles/year/category/database and acceptance prospects are not verified.** Final journal choice follows the Phase8 result, later metric/policy verification and author approval; no submission or contact occurs now.

17. Complete phase roadmap

0 planning (current) → 1 exact-gap/source/scope review → 2 inherited-input/local-form contract → 3 approved bounded pilot → 4 core theory → 5 sharpness/justified extensions → 6 independent certificates → 7 optional quantum → 8 independent mathematical/novelty audit → 9 new manuscript → 10 publication gate.

Every arrow requires a report and explicit next-phase approval. **No manuscript drafting before Phase8 audit and separate Phase9 approval.** 08 gives work packages, estimates, author dependencies, measurable exits and stop conditions.

18. Measurable exit criteria and present disposition

Blueprint delivery criteria

- Ten requested documents and a substantive18-topic coverage map.
- P1/P2/P3 comparison/scoring, with one advisory primary recommendation. -43 audit-family constraints and16 follow-up observations preserved/linked; no old closure state changed. -18 new author decisions, each with required choice, alternatives, recommendation, mathematical justification and consequences; all pending.
- Byte-exact source context, baseline/preservation evidence, valid local navigation, full file hashes, ZIP CRC/member/hash and standalone verifier checks.

These are **planning/integrity** checks. Their results are recorded in the delivery validation and verifier output, not treated as mathematical GO.

Future research/paper criteria

A defensible bounded gap; valid premises; complete independent proof review of a nonroutine structural theorem plus consequence; exact named-object certificates; no critical unresolved dependency; explicit prior-work comparison; QG1–QG6 if quantum retained; author decisions and journal/policy facts confirmed. Otherwise restrict, defer or stop.

Topic coverage map

Required topic	Principal detailed location
1 Title	03 §1; master §1
2 Motivation	03 §2; master §2
3 Verified literature	02 §§1–3; evidence web/source ledgers
4 Precise gap	02 §4; 03 §§3–4
5 Objective	03 §3
6 Research questions	03 §§4–5
7 Algebra/assumptions	04 §§1–2
8 Definitions/framework	04 §§3–6
9 Theorem roadmap	05 §§2–4
10 Expected contributions	03 §§6–7; 05 §4
11 Computation	06 §§1–10
12 Quantum	07 §§1–7
13 Examples/parameter tables	06 §§4–6
14 Novelty comparisons	01–02; 05 target-specific tests
15 Risks/alternatives	03 §8; 09 decisions
16 Q1-journal positioning	08 §4
17 Complete phase roadmap	08 §§2–3
18 Measurable exits	08 §3; master §18

Evidence, preservation and use

- Source study, authority and limitations.
- Source context index and 79-file identity manifest.
- All43 research-constraint mappings.
- 16 inherited follow-up mappings.
- Pre-task2,388-file preservation baseline.
- Final original preservation and delivery validation.

Selected historical context copies are **uncorrected partial trees**. Their original links/proofs/claims are not automatically endorsed or made standalone. The new blueprint's own links and supporting context destinations are checked. Raw external PDFs/HTML and large scientific datasets are not newly generated or bundled as new evidence.

Date provenance: the baseline workspace clock records **2026-11-13T10:20:03.088474Z**; its raw `session_declared_date` says **2026-11-12**. That field is **not a verified execution date**, and the earlier unsupported planning-date interpretation is withdrawn in PROVENANCE_CORRECTIONS.md. No old date or seal is rewritten. Integrity verifies files, not scientific truth, priority, quartiles or authority.

The root deliverable is **NEW_RESEARCH_BLUEPRINT.zip** with its external **.zip.sha256**. The README explains read-only verification. The only sensible next action is author review of NAD01–NAD18—not automatic proof development.

FINAL STATUS: BLUEPRINT DEVELOPMENT ONLY. NO THEOREM PROVING OR RESEARCH IMPLEMENTATION AUTHORIZED. WAIT FOR EXPLICIT AUTHOR APPROVAL BEFORE PHASE1.

Author final scope decision — proposed freeze, awaiting author choice

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BLUEPRINT REVIEW ONLY. This is a decision form, not a signed scope freeze or Phase 1 authorization.

All ten numbered blueprint documents and all 18 NAD entries were reviewed against the 43-family constraint crosswalk. **Every author choice remains PENDING / NOT RECORDED.** Arena's recommendations below do not approve themselves, adopt definitions, establish novelty or close audit findings.

Executive summary — minimum response before Phase 1

The recommended **review-only scope freeze** needs four explicit confirmations, covering six NAD entries:

1. **Direction — NAD01:** P2-S, P1, P3, or defer.
2. **Review scope — NAD03–NAD05:** proposed Core A, matrix size 2, all twists, using the provisional conventions in §§3–4; or specify changes.
3. **Prior-source treatment — NAD02:** confirm the exact source and factual editorial status; identify a later version; or explicitly leave them unknown and use MS03 as a **provisional staging reference only**.
4. **Authority — NAD18:** approve **Phase 1 only: primary-literature, source-authority and scope review**, or request revisions/hold.

Arena recommends: P2-S; Core A/all twists; factual source confirmation where available and an explicit uncertainty flag otherwise; bounded Phase 1-only approval **if the author agrees**. No such approval is recorded here. One clear author response can cover all four confirmations; eighteen separate approvals are unnecessary.

The old-paper freeze and no-Git/publication boundary already bind this task. Remaining decisions may be deferred as specified in §10, not silently decided. Unknown source facts need not prevent limited source/literature review, but they block a definitive old/new overlap or priority conclusion. Blank fields, agreement with a title, or receipt of this file are not authorization.

1. Recommended research title

Generator Profiles of Galois Hulls of Principal Cyclic Codes over Matrix Rings

This working title assumes neither generator growth nor a quantum application. A narrower eventual title must follow the surviving theorem. **Author's final title: PENDING.**

2. Primary problem and reassessment of P2-S

For **principal** matrix-ring cyclic inputs, seek a structural description of hull generator profiles: when does the hull remain principal, and, if it does not, what pairing/embedding mechanism permits a nonprincipal hull? An internal open question, MS03 O-3.1b, motivates the proposal; **it is not an external novelty certificate**.

Direction	Existing advisory score /100	Reassessment from the completed blueprint
P2-S	**73.6**	Best bounded starting point: principal-input structure, rather than known nonprincipal-input witnesses. Main risk: scalar hull classification, MT/intersection generators or Artinian theory may already imply the answer routinely.
P3	65.6	Reserve alternative. Requires a genuinely new, well-defined Ore quotient/adjunction compatibility result; skew + Galois and matrix/form transport are occupied ingredients.
P1	50.8	Reserve alternative. Needs a new trace-compatible physical mechanism, not the inherited double-twist identity, R1/EA routes or another q8 example.

These are the original **subjective planning scores**, not fresh measurements or probabilities. No new literature search or rescoring was performed for this review. P2-S falls to **60.8** under the blueprint's novelty/headroom sensitivity downgrade, below P3. Thus its recommendation is conditional: if Phase 1 finds only known/routine content, **stop or return a revised proposal to the author**; do not automatically switch direction.

Arena recommendation: P2-S for the gap-review gate only. **Author's final direction: PENDING.** See comparison and novelty positioning.

3. Exact proposed mathematical scope — review target, not adopted theory

Item	Proposed Core A boundary
Parameters	p=2; q=2^m with m≥2; matrix size r=2; nilpotency t=2; cyclic length n=2; 0≤s<m
Alphabet and ambient ring	A=F_q[u]/(u²); R=M₂(A); X central; B=A[X]/(X²−1); S=R[X]/(X²−1)≅M₂(B)
Inputs	Principal **right** ideals C=gS, including the zero case under μ(0)=0; no freeness or full-rank assumption on g
Morita convention	Embedded **column** module W=im_B(g)⊆B², with μ_B(W)≤2; r=2 denotes matrix size, not rank(g)
Twist and dual	σ_s acts on coefficients by a↦a^(2^s), fixing u,X; θ_s=σ_s∘transpose; first-slot form B_s(x,y)=Σ_i θ_s(x_i)y_i on R²; C^(⊥s)={y:B_s(x,y)=0 for all x∈C}
Hull and generator question	Hull_s(C)=C∩C^(⊥s); H_s(W) denotes its actual Morita module. The principal-input question asks whether μ_B(H_s(W)) can exceed 2, not merely whether μ increases.
Proposed local model	T=F_q[u,w]/(u²,w²), w=X−1, as in blueprint 04; its duality must be **induced from the global form**, not replaced by a convenient dot product on T².

Not in the initial research scope: r≥3, arbitrary t/e, odd-characteristic extension, multiple CRT factors, full wild-module classification, skew variables, quantum constructions or scientific computation. Related literature may of course be compared without adopting its broader research scope.

Core B (a specified two-generated class with e=p^a>2) and Core C (nonroutine coupled-CRT synthesis) remain **separate future scope decisions**, not fallback permissions. The q4 census is inherited finite evidence; it does not establish a theorem for every m,s. See problem and framework.

4. Ring and module assumptions requiring confirmation

Approval of the review target should explicitly address:

- **Side/slot:** right cyclic ideals, column Morita convention and the stated first-slot θ_s dual. A different side or slot needs a translated source comparison, not a cosmetic substitution.
- **Automorphisms:** coefficient Frobenius only, with u,X fixed; no moving-u automorphisms or Ore relation are included.
- **Non-reflexivity:** all twists are questions to review, not assumed involutions. The inherited primary convention gives DD=+σ_(2s)(C); universal reflexivity and codewise invariance are distinct.
- **Scalar level:** b_s is matrix trace followed by coefficient of u, valued in F_q, not the absolute F_2 trace. Ring/scalar orthogonal equality needs its valid module/ideal-catching hypotheses.
- **Embedding/equivalence:** no freeness assumption; no identification of μ with dimension or socle type; no arbitrary ambient GL change treated as a hull-preserving isometry.

The **complete meaning of "generator profile", sufficient invariants, allowed normal-form equivalences, exact local functional and coefficient-field/coordinate maps remain unsettled**. Layer dimensions, generator relations and pairing/embedding data are candidate descriptors, not an approved complete invariant. Explicit field encodings are deferred under NAD14.

This is a proposed comparison contract. Author agreement to review it does **not** certify inherited proofs, finalize those outstanding definitions, or authorize their development. Final applicability/local-form specifications belong to a separately approved Phase 2.

5. Proposed generator-profile questions

1. **Preservation versus nonprincipality (RQ1/RQ3):** within Core A, can μ_B(W)≤2 coexist with μ_B(H_s(W))>2, or is preservation forced?
2. **Structural sufficiency (RQ2/RQ4):** which generator relations, nilpotent layers and embedding/form data actually determine μ and the retained hull profile? Coarse dimensions are not assumed sufficient.
3. **Sharp range (RQ5):** what bounds, attainable profiles or preservation classification hold in the retained class? No positive-growth family is promised.
4. **Twist dependence (RQ6):** what changes with s, including m∤2s, without rebranding the inherited DD identity as new?
5. **Conditional consequences (RQ7/RQ8):** only after renewed approval, does a result support nonroutine CRT synthesis or a certificate-producing structural algorithm?

Arena recommends question-only posture. PC1 remains an unadopted, unproved exploratory conjecture; old H-3A.1/H-3A.2 are not premises. Formal conjecture adoption is NAD16, not implied by Phase 1 review.

6. Expected theorem-level contributions — targets, not results

The intended core is **NT03–NT05**: a nonroutine structural profile theorem, a principal-input criterion/preservation-or-growth classification, and a meaningful sharpness or attainable-spectrum consequence. NT06 twist/embedding dependence is conditional; NT07–NT08 extensions/algorithms need further scope and execution approval.

Paper-depth gate: at least one substantial structural main theorem plus a meaningful proved consequence/classification must survive independent proof and predecessor review. A preservation theorem is acceptable if substantial; a known/routine answer or isolated counterexample may resolve a question without supporting the intended paper.

NT01–NT02 are source-validity/local-form infrastructure, normally not novelty. Standard Morita packing, Nakayama, CRT, a generic nullspace/GPM computation, a parameter table or repaired old proof is insufficient alone. **No theorem work is authorized here or by a Phase 1-only approval.** See roadmap.

7. Explicit exclusions from MS03 and the inherited audit

Do not present the existing matrix/Galois/Gray/quantum chain, DD formula, principality threshold, chain/non-chain boundary, generic hull intersection, valid nonprincipal-input witnesses or existing quantum records as this project's new contributions. In particular:

- Do not repair/relabel the failed generic μ3 family; the audited family has μ2. Separate valid witnesses do not repair its proof.
- Do not confuse a nonprincipal **dual** of a principal input with a nonprincipal **hull** of that input.
- No manuscript editing/building, old-script repair, rerun, data replacement, benchmark installation or correction authorization follows from this scope review.
- No “first q8 EA”, “first matrix-ring quantum”, generic new Singleton-bound or rank-substitution novelty claim.
- Retain E/U/PC/RQ/X and finite-evidence distinctions; printed-proof validity, theorem truth, numerical certification and priority remain separate.

The old project stays **CONDITIONAL HOLD**. Its 20 author decisions and all 43 correction families remain open/unimplemented; this review cannot close them.

8. Literature verification requirements and Phase 1 exit

Present status: external gap X — unverified. A later authorized Phase 1 must compare exact primary statements, not infer absence from previews or failed searches:

- MS03/later author-identified source; foundational anti-isomorphism/Frobenius duality and relevant module algebra.
- NL01–NL02 serial/repeated-root theory; **NL03–NL04** MT/intersection generators; **NL10** scalar repeated-root hulls; **NL11** Artinian μ/type/duality and their decisive predecessors.
- NL05–NL06 for the P3 comparison; NL07–NL09 and inherited A8/Cao/C5/C6 if a skew/quantum expansion is later proposed.

For each core target NT03/NT04/NT05 record exact hypotheses, side/slot, input class, output invariant, theorem/version locator, access level and whether it is already a routine specialization. Preserve unresolved MT2025/2026 relationships, A5a author order, A11 metadata and preview-limited cells; do not invent resolutions.

Exit: at least 12 substantive source/theorem comparison rows covering the required groups; a direct comparison for each core target or an explicit blocked/X label; a source/overlap boundary; scope revision/stop recommendation; author confirmation of acceptable outcomes and the independent-review route before seeking Phase 2 approval. A row count alone is not sufficient. Unknown baseline facts or inaccessible load-bearing texts prevent the affected affirmative gap conclusion.

Phase 1 may end **GO-for-consideration / NARROW / BLOCKED / STOP**, followed by an author decision. It does not prove a new theorem, establish all-literature absence, certify Q1 status or start Phase 2 automatically. This review performed **no fresh primary-literature verification or Phase 1 execution**.

9. Optional quantum extension and approval gate

Arena recommendation (NAD07): algebra first; quantum conditional, or omit it. **Author's final quantum choice: PENDING.** It is not part of Phase 1 execution and is not required to rescue a weak algebraic result.

Later separate author authority and all QG1–QG6 are required: a genuine structural consequence; the correct form/field bridge; metric preservation or physical-coordinate distance recertification; trace isotropy and phase/scalar-free lift or valid Bob completion; exact named-object certification; and incremental novelty beyond standard EA/CSS formulas.

μ_B(H_s), transported h_s, ordinary h_E and a trace commutator radical are **not interchangeable**. SO/DC/EA/R1 use different inputs, orthogonals and logical-error sets. Match physical alphabets/resources and exact-versus-bound distances. Failure of the bridge means **omit/defer**, not assume Gray compatibility or discard an audit condition.

10. Author decisions and proposed freeze record

A. Essential entry decisions — six NAD entries, four response blocks

Every "Author choice" field below is PENDING. Option letters retain their meanings in register 09.

NAD01 — Direction. A: P2-S → bounded principal-input structural review, novelty still open. B: P1 → new physical-mechanism comparison needed. C: P3 → new quotient/adjunction contract needed. D: defer → no Phase 1.

Arena: A, because the internal question is sharper and more manageable, not because literature absence is known. **Author choice: PENDING.** Selecting B/C requires a revised review scope before execution; the P2-S assumptions do not carry over automatically.

NAD02 — Prior source/status. A: confirm MS03's exact identity **and actual editorial facts** → meaningful baseline. B: identify a later version → re-inventory and reassess overlap. C: explicitly unknown → MS03 staging reference only; definitive overlap/submission/priority claims blocked.

Arena: factual confirmation; C if facts are unavailable, rather than invent them. Baseline identity affects whether a proposed contribution already exists internally. **Author choice/facts: PENDING.** Preparation folders are not submission evidence.

NAD03 — Matrix size. A: r=2 → focused principal-input class. B: r=3 → larger scope; nonprincipality requires μ>3, not μ3, and rank substitution is insufficient novelty. C: all r → larger classification burden and packing-threshold overlap risk.

Arena: A, because the structural question, not matrix enlargement, should supply content. **Author choice: PENDING.**

NAD04 — Initial family. A: Core A in §3 → precise small starting class, possibly routine. B: specified two-generated t=2,e>2 class → richer but requires named restrictions/transport. C: arbitrary t,e/modules → full-wild complexity and poor initial feasibility.

Arena: A; B only after an explicit revised scope, not automatic escalation if A is shallow. **Author choice: PENDING.**

NAD05 — Twists. A: all s → review non-reflexive as well as reflexive cases. B: m|2s only → simpler symmetry, weaker Galois scope. C: s=0 only → bilinear control, not an all-twist contribution.

Arena: A as a question range; exact side/form hypotheses must survive review. **Author choice: PENDING.**

NAD18 — Phase authority. A: bounded Phase 1-only review → source/gap/scope report, then stop. B: blueprint revisions → planning only. C: hold/stop → no research execution.

Arena: A only if the author explicitly accepts the selected review target and safeguards. The unresolved gap is why review must precede proof/code. **Author choice: PENDING.** No proof, research implementation/experiment, manuscript edit/build, old correction, Git write, release, deposit, editorial contact or submission is included.

B. Other twelve decisions — retained, not obstacles manufactured at entry

The full alternatives/justifications in register 09 remain controlling. Below are their timing and consequences for this proposed freeze. **All final author choices remain PENDING.**

NAD / timing	Alternatives and consequences	Arena recommendation / rationale	Author choice
NAD06 — before Phase 1 exit / theory scope	A outcome-neutral → preservation or growth; B growth subquestion only → negative outcome may end project; C defer → no theory programme.	A; truth cannot be chosen, and a routine answer triggers re-scope/stop.	PENDING
NAD07 — optional track	A conditional quantum → bridge gates; B algebra-only → no physical claims; C quantum-led → revised plan.	A conditionally; B if no useful bridge. μ is not entanglement cost.	PENDING
NAD08 — binding boundary now	A freeze old work; B request separate correction task; C seek source-backed clarification. B/C do not authorize corrections here.	A; old proof repair is not new research. Current preservation instruction already binds regardless of a future vote.	PENDING
NAD09 — before a positive gap verdict	A primary texts → exact comparison; B previews only → dependent claims stay X; C defer inaccessible dependencies → blocked verdict.	A with C where necessary; access/provider choices need not all be settled before source review starts.	PENDING
NAD10 — before any pilot	A bounded proposed pilot; B no science computation; C custom cases/cap → new preflight/approval.	No computation in Phase 1; consider A only later. The old proposed budget is not permission.	PENDING
NAD11 — route by Phase 1 exit; reviewers before later audit	A independent proof/certificate review; B self-review only → lower assurance, not independent validation; C defer publication.	A, C if unavailable; shared code agreement is insufficient. No verifier implementation now.	PENDING
NAD12 — journal selection deferred	A provisional DCC; B conditional FFA; C defer final selection. Neither scope nor metrics certify Q1.	Keep DCC only as a scope lead; defer final choice until results/metric verification.	PENDING
NAD13 — before data generation/release as applicable	A immutable certificates/approved storage and licence; B private initially → disclose limits; C defer → no deposit/licence/public claim.	Plan A later; defer release now. Provenance and access/licence are distinct.	PENDING
NAD14 — before coordinates/implementation	A preserve inherited encodings; B checked isomorphisms → recertification duties; C defer implementation.	A for reused matrices; no numeric encoding finalized now. Isomorphic fields need not share integer labels.	PENDING
NAD15 — overlap boundary in Phase 1; declarations before drafting/publication	A accurate own-work disclosure; B wait for facts; C revise scope → remove duplication, not hide it.	A with B for unknown facts; final authorship/declarations need not be invented to start literature review.	PENDING

NAD / timing	Alternatives and consequences	Arena recommendation / rationale	Author choice
NAD16 — before formal conjecture adoption	A question-only; B retain exploratory PC1 → still unproved; C defer formulation until approved pilot.	A; absence of a q4 counterexample is not a universal proof. No conjecture is assumed while undecided.	PENDING
NAD17 — binding boundary now	A preserve/no Git/publication action; B later bounded Git request; C later release/editorial request. B/C need separate authority.	A now; no change to old seals or publication state.	PENDING

Entry versus exit: source-access, negative-outcome and review-route decisions need resolution or explicit blocked dispositions at the Phase 1 exit gate; they do not require selecting computational tools, reviewers, journals or a formal conjecture before a strictly bounded literature/source review. Deferral never waives validity, disclosure or permission requirements.

C. Consistency check — all 43 consolidated constraints retained

The 43-family crosswalk was checked against the consolidated source records: identities, severities and open/unimplemented dispositions agree. The table below covers every MC family once. **No scope-level conflict was identified in the recommended review-only package; no mathematical closure is claimed.** All eight critical families—MC10, MC23, MC24, MC25, MC27, MC29, MC32, MC40—remain unresolved.

Families	Safeguard carried into this proposed freeze
MC01–MC06, MC12	Right action/slot, positive DD and exact scalar/ideal-catching hypotheses; no invalid failure example, unestablished alternative pairing, blanket reflexivity or Hom shortcut.
MC07–MC10	Base/cyclic rings and cyclic/principal distinctions; inherited principality boundary is not new; generation is not freeness; failed μ3 family is not imported.
MC11, MC18–MC21, MC30–MC31	Correct Morita/image representation, non-tautological diagnostics, literal finite/sample/method scopes, actual field labels and named-object certificates; no campaign rerun or certificate transfer. These remain later computational gates.
MC13–MC17	Bilinear/Hermitian/Galois form and characteristic distinctions, actual GF9 transformation, code-specific CSS versus ambient obstruction, and F_q versus absolute-trace codomains.
MC22–MC29	Quantum phase/lift/isotropy, Bob completion, actual trace/F_q orthogonal, h_E, route-correct relative sets/coset metrics and bookkeeping. Quantum deferral **does not retire** these constraints; 742 metadata failures are not 742 false distances.
MC32–MC33	Physical-alphabet/resource/header matching; no global table-absence claim from a missing endpoint. No benchmark claim or reconstruction now.
MC34–MC36	Correct source identities and unresolved metadata; exact predecessor/priority comparisons, including A8; no new-bound/firstness/Q1 guarantee.
MC37–MC39	No first-q8 P1, parameter-only P2 or ingredient-only P3; matrix thresholds/index/cost and Ore compatibility remain correct.
MC40, MC42	Staging identity is not submitted/current authority; retain author-only gates, unavailable-PDF/reference-closure limits and separate drafting approval.
MC41, MC43	Preserve counting bases, execution-attribution/date discrepancies, source bytes and completed seals; no retrospective execution claim or gratuitous rerun.

All 43 remain **OPEN / AUTHOR CHOICE AND REVIEW REQUIRED; NOT AUTHORIZED / NOT IMPLEMENTED** (8 critical, 29 major, 6 minor). The linked 16 RF observations, 30 conflicts and old 20 author choices are not removed or resolved. Counts are issue/evidence classifications, not counts of false theorems. Changed alternatives must be rechecked against these constraints before use.

D. Author response record — unfilled

Minimum response block	Author's final response
Direction (NAD01)	**PENDING — NOT RECORDED**
Review scope, assumptions/changes (NAD03–NAD05; §§3–4)	**PENDING — NOT RECORDED**
Exact prior source/editorial facts, or explicit unknown treatment (NAD02)	**PENDING — NOT RECORDED**
Phase 1-only approval / revisions / hold (NAD18)	**PENDING — NOT RECORDED**

A future approval should state that **all 43 constraints and the no-proof/no-science/no-manuscript/no-Git/publication boundary remain in force**, with no automatic next phase. Other NAD choices stay pending unless expressly answered. Disputed facts or a changed direction/scope must be stated, not inferred.

Review basis: blueprint 01, 02, 03, 04, 05, 06, 07, 08, 09, 10, the 18-decision index and the complete constraint crosswalk. This is a scope/decision review, not a new proof or independent technical re-audit.

Preservation: this new addendum is outside the previously sealed ZIP and its 109-file member set. The ZIP, checksum, original ten documents, decision/evidence registers and `SHA256SUMS.txt` are **not rewritten or resealed**. The old manifest does not purport to hash this later addendum. Frozen ZIP SHA-256: `23dcc025177ddbf917abbd00c0d381a2e70d15af56eeb53b11c31fa098e6b0b6`.

FINAL STATUS: BLUEPRINT REVIEW ONLY. NO PHASE 1 EXECUTION. WAIT FOR THE AUTHOR'S EXPLICIT DECISION.

01 — Candidate research directions: comparison and recommendation

SOURCE / 01_CANDIDATE_DIRECTION_COMPARISON.md

Status: BLUEPRINT DEVELOPMENT ONLY. No direction is author-selected.
Arena recommendation: a narrowed structural P2, not a size-three extension.

Master blueprint · Literature · Author choices

1. Non-negotiable starting point

MS03 already contains the matrix-size-two cyclic/Galois-duality/hull/Gray/quantum chain. Its audited positive double twist, Morita principality criterion and valid finite nonprincipal hull witnesses are **prior internal work**, not proposed new contributions. Its defective printed proofs and unsupported comparisons do not become new results when corrected. The old project remains **CONDITIONAL HOLD**; the new proposal neither edits it nor closes its43 issue families or20 decisions.

The new paper needs a **different theorem-level question**. The 43-family constraint crosswalk and source study govern all candidates.

2. Mathematical comparison

Dimension	P1 — non-reflexive Galois duality and quantum constructions	P2 — structural Galois hulls and generator numbers	P3 — skew-cyclic × Galois duality over matrix rings
Distinct research question	Can a genuinely twist-dependent, trace-compatible stabilizer/EA mechanism be characterized outside involutive duality?	Which **principal inputs** have principal/nonprincipal hulls, and what embedding/module data determine hull generator profiles?	Which nontrivial Ore automorp hism/anti-automorphism compatibilities give a new duality/hull theorem not obtained by known transport?
Already in MS03	DD=+σ_(2s)(C), non-reflexive examples, hull self-orthogonality, R1 and EA routes, **four q8 EA records**	Morita intersection formula; μ-to-principality threshold; chain/non-chain boundary; μ=3 hulls, including proper hulls of nonprincipal inputs	Only the **central-variable** cyclic setting; not an established skew quotient
Genuine difference required	A new physical commutator/radical mechanism, not another q8 example or the DD identity	A structural resolution of O-3.1b or a substantial generator-profile classification/bound for two-generated Morita inputs	A valid quotient, sided adjunction and induced hull structure with a demonstrably nonroutine compatibility obstruction/construction
Nearest occupied terrain	Field Galois hull→EA, hull decompositions, trace/additive stabilizers, simultaneous-Galois sampling	Scalar serial hulls; **repeated-root scalar hulls**; MT/QC hull generators; Artinian μ/type/dual invariants	**Published skew MT Galois duals (2019)**; γ-duals/skew chain rings; matrix-ring skew codes; Ore→matrix/form transport
Possible theorem package	Trace-radical characterization, twist-sensitive construction with correct distance, impossibility/attainability theorem	Generator-profile theorem, principal-input dichotomy/criterion, sharpness or preservation classification, justified certificate algorithm	Quotient/adjunction compatibility theorem, dual/hull closure criterion, nontrivial obstruction or classification, metric-qualified images
Feasibility	Medium-low: a physically meaningful new bridge may disappear after correct Fp/Fq translation	Best if restricted first to rank2, t=2, repeated-root local factors; full wild-module classification is not proposed	Lowest: well-definedness precedes structural work; a formal tensor/matrix extension may add no new theorem
Computation needed later	Independent finite-field and Fp commutator checks; phase/Bob checks; relative-distance certification	Small exact module/radical computations and normal-form-directed examples; full high-q subspace enumeration avoided	Ore multiplication, ideal-sidedness, adjunction and negative controls before distance work
Quantum relevance	Direct, but unestablished as a new mechanism	Optional application only; μ(hull) is not h_E or entanglement cost	Optional and high-risk; skew+Galois closure does not imply physical isotropy
Q1 aspiration	Conditional on a new theorem, not a first-alphabet application	Conditional on an explanatory structural result beyond Nakayama/Morita and existing GPM algorithms	Conditional on substantial new compatibility/structure, not combining keywords
Main failure mode	Relabeling old R1/EA or using the wrong orthogonal	Reproving known generator formulas or treating an internal open question as an external gap	Repackaging known Ore/matrix equivalences, invalid two-sided quotient, or incorrect side/form transfer

Literature evidence levels and exact identities are in 02 and fresh retrieval notes. Positive reported scope is distinguished from independently checked proof applicability.

3. Comparative scoring matrix

Scale1–5, where5 is **more favorable**. These are reasoned planning judgments, **not measured novelty, acceptance probabilities, verified quartile scores or author choices**. Scores refer to the differentiated versions above, not to disallowed ingredient-only variants. Literature and proof gaps remain open.

Criterion	Weight	P1	P2 narrowed	P3
Potential mathematical novelty, if gap survives	18	2	4	4
Distance from MS03's actual contributions	14	2	4	4
Literature headroom, after known overlap	14	2	3	2
Potential depth of theorem-level contribution	14	3	4	5
Feasibility within a bounded programme	12	3	4	2
Computational tractability	8	3	4	2
Justifiable quantum application potential	6	4	2	3
Prospective fit to a strong mathematical coding journal	8	3	4	4
Risk manageability	6	2	3	2
Weighted total /100	**100**	**50.8**	**73.6**	**65.6**

Calculation: Σ(weight×score)/5. P2 wins because a concrete unresolved internal question and known algebraic machinery give a bounded starting point; it does **not** win because the literature is certified empty. Downgrading P2's novelty from4 to2 and its literature headroom from3 to1 reduces it to60.8, below P3. The recommendation must therefore be revisited if Phase1 finds an existing structural theorem that settles the proposed gap.

4. Recommended primary problem — provisional P2-S

Working problem: determine the generator profiles of Galois hulls of principal cyclic codes over $M_2(F_q[u]/u^2)$, beginning with a precisely transported repeated-root local model and retaining the embedding data that plain dimensions forget.

The principal-input question is sharper than “nonprincipal hulls exist”: the known μ=3 hulls do not answer it because their input codes are already nonprincipal. The existing1,221-module/2,442-hull audit checks only the specific (q,t,n)=(4,2,2) family; it is not a general theorem over all fields or multiplicities. The problem specification treats either a principality-preservation theorem or a structurally classified growth mechanism as a legitimate outcome. No positive answer is assumed.

Minimum hoped-for advance: a criterion/bound/classification using input module type **and its pairing/embedding data**, with sharpness or a principality-preservation theorem. Merely computing μ of an already computed hull by Nakayama, adding a finite witness, or renaming a QC hull algorithm is insufficient for the intended paper.

5. Disallowed shortcuts and resource comparison

- **P1:** “first q8 EA” is contradicted by included records. DD=+σ_(2s) is inherited. R1 must use its actual generally Fp-linear label group and trace orthogonal.
- **P2:** changing r=2 to r=3 alone is not a research gap. In the general known threshold, principal iff μ_B(W)≤r; at r=3 a nonprincipal example needs μ>3, not merely3. Coefficient-expansion index is tr^2, hence9t; for $q=p^2$,t=2,r=3 the **base ring** has p^36 elements, and length-n ambient has p^(36n). Those are cost corrections, not new theorems.
- **P3:** field skew+Galois and matrix/chain-ring skew precedents must be compared. Two-sidedness of (X^n−1), the automorphism order, module side, adjunction and shift compatibility cannot be inferred from the central-variable case.
- A fourth direction is **not recommended now**: simultaneous-Galois hull algorithms would face NL09's reported sampling/type work and would require its own gap review. It is not a hidden substitute approved by this blueprint.

6. Decision boundary

AUTHOR DECISION NAD01 in 09 compares P2-S/P1/P3/stop with mathematical justification and consequences. **NAD18** controls the first future phase. This document recommends P2-S for author consideration only. No theorem is proved, no conjecture adopted and no scientific implementation begun.

BLUEPRINT DEVELOPMENT ONLY — wait for explicit author approval before Phase1.

02 — Literature evidence and provisional novelty positioning

SOURCE / 02_LITERATURE_AND_NOVELTY_POSITIONING.md

Status: BLUEPRINT DEVELOPMENT ONLY. External nonoccupation is not certified.

Master · Comparison · Evidence notes

1. Evidence rules

Label	Meaning here	Permitted inference
E — established background	Standard published mathematics or an inherited audited conditional result, with stated hypotheses	May be a cited input after applicability review; not a new contribution
U — unresolved result	Existing manuscript/register question or a remaining proof/applicability issue	Not a theorem; old computational evidence stays bounded
PC — proposed conjecture	Explicitly tentative future proposition, not used as an assumption	May guide later authorized falsification; no present proof claim
RQ — new research question	The differentiated question proposed for a new project	New to this proposal, not certified first in the literature
X — externally unverified	Priority, exact gap, all-version absence or incompletely checked applicability	Must be qualified; cannot appear as an established novelty claim

A publisher abstract verifies what a paper **reports**, not its complete proof. An authentic PDF does not validate every ring identification it prints. A dash or no-match search does not prove absence. The old37-entry bibliography and the audited **38 source rows/266 cells** are different sets.

2. Mandatory inherited literature constraints

The complete source-specific dispositions are preserved in the source ledger, cell ledger and master literature blueprint. The new programme must not revert to the old literature matrix's unsupported negative cells.

Predecessor	Verified identity / reported scope retained	Consequence for this proposal
Wood1999; Wood2010; Szabo–Wood2017; Dougherty–Leroy2016	Frobenius/character duality, anti-isomorphisms, nondegenerate forms and noncommutative self-duality; member-specific hypotheses	Transpose/anti-automorphism duality is not new by naming it. Mixed left/right dual compositions must not be identified with identical-twist DD
Fan–Zhang2017; Liu–Pan2020	Field Galois duality/hull dimension methods; Liu–Pan DOI10.1007/s10623-019-00681-2	Generic twisted Gram-rank formulas are background; field matrix-product codes are not automatically matrix-ring alphabets

Predecessor	Verified identity / reported scope retained	Consequence for this proposal
Cao2021, IEEE TIT67(12),7964–7984	GRS/extended-GRS Galois hulls and EA applications, DOI10.1109/TIT.2021.3117562; reported application restriction2ℓ divides e	P1 must not claim an unrestricted hull-to-EA bridge without its own proof
Matrix cyclic/Gray/dual papers A1–A11	Several matrix sizes, nilpotent coefficients, Euclidean/Hermitian duals and Gray maps, with prime/length restrictions	Larger r, nilpotency or a Gray map is not itself a gap; preserve alphabet and version distinctions
A8 Joshi–Patel–Prakash, QIP25:207(2026)	DOI10.1007/s11128-026-05230-0; matrix cyclic→special Gray→CSS, including F27 applications	**“First matrix-ring quantum construction” is unavailable**
C5 Zhang–Kong–Zheng, Entropy28(4):407(2026)	DOI10.3390/e28040407; non-chain commutative Galois hulls, EA and restricted ordinary-QECC applications	Do not assume all twists or all Gray images are physically compatible
C6 Debnath–Islam–Martínez-Moro–Prakash, DM349(2):114750(2026)	DOI10.1016/j.disc.2025.114750; commutative affine algebra, explicit Gray/hull transport and EA	A generic Gray/hull/EA combination is occupied
Kushwaha–Prakash C7a/C7b	Distinct free-hull and symplectic left/right/two-sided hull preprints	“Noncommutative hulls did not exist” is false as a broad positioning claim
Calderbank–Shor, Steane, Gottesman, Ketkar et al.; Brun–Devetak–Hsieh, Wilde–Brun, Lai–Brun–Wilde	Stabilizer/CSS/EA foundations, member-specific source roles	Standard construction formulas are imports, not expected new theorems

Residuals **remain**: A11 venue/year/DOI; A5a final author order; preview-limited technical cells and NRS chapter attribution. Correct A1 DOI ends **.023**, not .008. Fan–Zhang's volume is84, already correct in the BibTeX. These facts are inherited, not freshly reverified metadata for every source.

3. Fresh nearest-neighbour checks

The web notes give exact titles, URLs, version conflicts, inspected chunks and failures. The following compact matrix is not a full-paper proof audit.

ID / source	Positive scope verified at the recorded level	What it rules out as novelty	What remains to compare
NL01 Talbi et al., FFA77:101950(2022), DOI, preprint	Galois hulls of cyclic **serial** chain-ring codes; gcd(n,p)=1 explicitly in §3	First Galois hull theory over chain rings; generic twisted duality	Rank-two repeated-root module hulls, exact conventions and generator profiles
NL02 Sălăgean, DAM1 54(2),413–419(2006), publisher	Repeated-root cyclic/negacyclic structure; cyclic codes generally nonprincipal; Gröbner/generator predecessors	First repeated-root nonprincipality or generic generator algorithm	Hull-specific constraints on two-generated embedded modules
NL03 Taki Eldin–Solé, FFA109:102712(2026), publisher, preprint	MT Galois hull dimension and **generator polynomial matrices**; Theorem18; compatible shift constants; QC special cases	Generic Galois hull generators, field QC/MT reformulations	Whether the proposed nilpotent/module-action invariant is already recoverable routinely from those generators

ID / source	Positive scope verified at the recorded level	What it rules out as novelty	What remains to compare
NL04 Takieldin–Leroy, arXiv2503.24303	Linear/MT intersection generators and Galois-hull applications reported	A generic nullspace/intersection algorithm as a principal contribution	Additional structural compression and module generator information, if any
NL05 Sharma–Chauhan, FFA59,297–334(2019), DOI	**Skew multi-twisted codes and Galois duals**, generators, SO/LCD/selfdual criteria	Ingredient-only skew×Galois novelty	Genuine matrix-ring/Ore compatibility theorem not a formal extension
NL06 Caruso–Drain, arXiv2410.12340v2	Selfdual skew cyclic codes; separable Ore quotient decomposes into matrix algebras (Prop2.4); trace/coordinate orthogonals (Prop2.7); adjunction transport	Ore→matrix decomposition/form transport or enumeration alone	A new noninvolutive or nilpotent compatibility obstruction, with precise side/form maps
NL07 Dinh–Thi–Tansuchat, Axioms13(10):655(2024), primary	γ-duals and skew constacyclic chain-ring applications reported	A blanket claim that automorphism-dual/skew chain-ring theory is absent	Full §4 theorem hypotheses, not only search-indexed compatibility conditions
NL08 Li–Liu, arXiv2408.09334	s-Galois hull decomposition and EA applications reported	First Galois hull decomposition-to-EA idea	Non-reflexive applicability, commutator interpretation, proof and metric conventions
NL09 Albrecht–Benčina–Lai, arXiv2610.08664	Odd-characteristic self-orthogonal and total-Galois sampling, prescribed hull/type	Naming simultaneous Galois hull algorithms as a fresh idea	Relation to any optional P1/P4 mechanism; preprint status preserved
NL10 Cao–Cao–Fu, AA ECC32,459–493(2021), DOI, preprint	**Repeated-root scalar hulls** at length2n over F_(2^m)+uF_(2^m), n odd; selfdual classification/mass formulas	Claiming the entire repeated-root hull problem is untouched	Whether coupled rank-two modules/principal matrix inputs add a nonroutine theorem beyond scalar constituents
NL11 Camps-Moreno et al., arXiv2509.05850, publisher DOI	Artinian-code socle/type/μ/length, Frobenius type+dual-free-rank theorem, LCD freeness	Nakayama, Artinian invariants or a generic μ/dual relation relabelled as new	Exact hull generator spectrum under principal-input and embedding constraints; no nonoccupation inferred from partial reading

Version limits: the ACM2025 MT-hull record and FFA2026 record may be related; that relationship is unresolved, not silently merged or double-counted. Caruso's HTML heading date differs from its arXiv version-history date. NL11's publisher preview did not return final issue/pages/year, which are not invented. NL04's verified preprint is used rather than an inferred final citation. New notes are transcriptions, not raw downloaded papers.

4. Exact proposed gap — and what is not yet established

Internally verified motivation: MS03's `rem:o31b` and conclusion explicitly leave open whether a principal cyclic code can have a nonprincipal Galois hull in the repeated-root, nilpotent setting. The follow-up's old census confirms no such example in its one checked family; valid existing nonprincipal hulls do not settle the principal-input question.

RQ: characterize attainable hull generator profiles, and decide principal-input preservation versus generator growth, first in a restricted rank-two local repeated-root family, keeping the actual induced pairing and embedding information.

X: no external source found in this bounded review has been technically shown to settle that exact programme. This sentence is **not** a certification that none does. In particular:

1. A specialization of NL10's scalar classification may settle some cases.
2. NL03/NL04's generators plus the retained nilpotent action may already give the desired criterion by a short standard calculation.
3. NL11 or older commutative algebra may imply a general bound/preservation theorem.
4. A nonprincipal hull of a principal input might already be a known module-intersection example under another terminology.

Each possibility is a Phase1/2 stop-or-narrow test. Simply adding the word “Morita” or changing matrix size does not differentiate a result.

5. Theorem-level comparison protocol for the first authorized phase

For **every** proposed core target in 05, create a comparison row with:

- exact statement, quantifiers, field/ring, side, slot and twist convention;
- whether inputs are arbitrary modules, ideals, scalar cyclic codes or **principal matrix cyclic codes**;
- whether the output is field dimension, a generator matrix, **minimal B-generator number**, or a structural characterization;
- induced reversal/component permutation and nilpotent action;
- proof mechanism and whether the proposed result follows by a routine specialization;
- primary theorem/page/version locator, access level and unresolved applicability cells;
- separate assessments of **truth**, **printed proof**, **finite evidence**, **newness** and **priority**.

Required positive comparisons include the original MS03, foundational anti-isomorphism work, NL01–NL06, NL10–NL11, and any predecessor they cite that supplies the crucial μ/intersection mechanism. Quantum or P3 expansions additionally require NL07–NL09 and the retained matrix/Gray/EA predecessors. Abstract-only review is insufficient for an affirmative gap verdict. If a needed primary text cannot be obtained, retain X and postpone the dependent claim; do not simulate access.

6. Novelty language permitted now

Permitted: “We propose to investigate generator profiles of Galois hulls of principal matrix-ring cyclic codes, an internal open question requiring further external comparison.”

Not permitted: “first”, “the literature has no such result”, “Q1-ready”, “low novelty risk”, “new merely in origin”, “all predecessors lack Gray/quantum constructions”, or “a true corrected old bound is a new theorem”.

The new paper must cite and delimit MS03 honestly even if the old manuscript is unpublished or on HOLD. Actual publication/submission lineage is an author fact (NAD02); the blueprint cannot choose how much prior work to disclose by renaming it. No new manuscript text is drafted here.

7. Journal positioning boundary

DCC and FFA have relevant mathematical scopes; 08 gives conditional fit and rejection/stop criteria. Their current **Q1 quartile, database, metric year and subject category are not verified**. A suitable scope is not a quartile certificate, novelty certificate or promise of acceptance.

BLUEPRINT DEVELOPMENT ONLY — external priority and all research phases remain unapproved.

03 — Recommended research problem

SOURCE / 03_SELECTED_RESEARCH_PROBLEM.md

RECOMMENDED ONLY — AUTHOR SELECTION PENDING

The requested filename is not a selection record. **No primary problem has been formally chosen or approved.** See NAD01, NAD03–NAD07 and NAD18 in the decision register.

Master · Framework · Theorem targets

1. Working title

Generator Profiles of Galois Hulls of Principal Cyclic Codes over Matrix Rings

This neutral title does not presume that generator growth exists. If a preservation theorem is eventually the main result, the title should reflect it; if nonprincipal hulls are structurally classified, the title may then say so. Neither stronger title is adopted now.

2. Motivation and difference from MS03

A principal right ideal of $M_2(B)$ corresponds to a B-submodule $W \subseteq B^2$ with $\mu_B(W) \leq 2$. Its hull is controlled by an intersection involving the Galois/reversal dual module. Over the non-chain repeated-root factors, being a submodule of a two-generated module does **not** alone control generator number. This motivates asking what the **special dual-pairing relation**, rather than arbitrary intersection, forces.

MS03 already supplies the framework, the intersection reduction, the threshold and nonprincipal hull existence for nonprincipal inputs. Repeating these is not the new paper. Its failed direct-sum proof also cannot be turned into a new contribution by repair. The proposed advance is an **explanatory classification/bound/criterion for the principal-input subclass**, with dependence on embedding data and twist made visible.

The original1,221-module audit is a finite anchor at (4,2,2), not a theorem for all q. Scalar repeated-root hulls, MT hull generators and Artinian generator invariants are occupied literature (NL02, NL03, NL10, NL11). The external gap remains X, not established novelty.

3. Primary objective

Determine which hull generator profiles can occur for principal cyclic right ideals over $M_2(F_q[u]/u^2)$, first in a **bounded repeated-root local family**, and identify structural conditions that decide whether the hull remains principal. Seek a theorem with an interpretable input invariant/normal form, not only an algorithm that computes an already known intersection.

Recommended scope order, pending author choice:

1. **Core A:** q=2^m, m≥2, t=2, n=2, all s modulo m; local $B=F_q[u,X]/(u^2,(X-1)^2)$. The inherited q4 family is a regression reference; it is not a new census.
2. **Core B, only after a useful Core A theorem:** t=2 and one repeated-root factor of multiplicity e=p^a; a restricted family of two-generated embedded modules, not classification of every wild module. Odd-characteristic comparison may begin with p=3,e=3 if justified.
3. **Extension C, conditional:** several CRT factors and their Galois/reversal coupling, **only** if it produces a theorem beyond existing CRT/max-μ assembly.

Matrix rank stays2 initially. r=3 or all r is a later structural corollary/extension only if new content remains; parameter substitution is not the objective.

4. Research questions and evidence status

ID	Precise question	Status now	Why it is differentiated
RQ1	For μ_B(W)≤2, can μ_B(H_s(W)) exceed2, where H_s(W)=W(C∩C^(⊥s)) for C=I_W?	**U/RQ**, internal O-3.1b; external X	Restricts the input, unlike known nonprincipal hull examples
RQ2	What module/embedding/pairing data determine the attainable pairs (μ(W), μ(H_s(W))) and local hull profiles?	**RQ; X**	Requires structural information beyond field dimension or generic GPM construction
RQ3	Is principal-hull preservation forced in Core A for every field/twist, or is there a describable obstruction?	**RQ**, not a yes/no prediction	The old q4 census is insufficient for a universal field-parameter statement
RQ4	Which layer/extension data distinguish equal-dimension/equal-μ inputs with different hull generator behavior?	**RQ**; no example promised	Prevents claiming a dimension-only formula without testing embedding dependence
RQ5	If growth occurs, what is the sharp attainable generator bound in the retained class, and what realizes it? If not, what forces preservation?	**RQ**	Accepts a substantial negative/preservation theorem, not a forced positive example
RQ6	How do coefficient Frobenius and reversal affect these conditions, including m∤2s?	**RQ**	Non-reflexivity is a variable to study, not a new DD theorem
RQ7	Does a local classification survive or change under actual CRT coupling, rather than a naive product of independent hulls?	**RQ**, extension only	Must exceed the already known permuted-intersection and max-μ facts
RQ8	Can the structural result support a certificate-producing method meaningfully better than ambient enumeration, with a proved bound on the stated input class?	**RQ**, conditional	A new wrapper around row reduction is not enough
RQ9	Is there a proved, metric-qualified connection to an ordinary Euclidean/Hermitian/trace hull giving a useful quantum consequence?	**RQ**, optional	μ_B and transported h_s are not entanglement cost by definition

The notation H_s(W) means **the Morita module of the actual global hull**, not an unproved standard dot-product hull on a conveniently chosen local basis.

5. Proposed conjecture — deliberately not an assumption

PC1 — exploratory principal-hull preservation in Core A: for q=2^m, m≥2, t=2, n=2, every principal cyclic right ideal C of $M_2(F_q[u]/u^2)[X]/(X^2-1)$ has principal Hull_s(C), for every0≤s<m.

Status: PROPOSED CONJECTURE, NOT ADOPTED OR PROVED. Evidence is limited to the inherited q4 audit and the absence of a principal-input counterexample there. No new computation supports other q. This is a falsification target, not a premise of any theorem or an external novelty assertion. A single valid counterexample would reject PC1; a search without one would not prove it. The author may prefer question-only posture (NAD16; Arena recommends that posture until the gap check is complete).

No analogue for arbitrary t/e/r or physical quantum parameters is conjectured by this blueprint. Old H-3A.1/H-3A.2 remain historical unresolved claims, not freshly adopted premises.

6. What would count as a new paper-sized result?

At least one **core structural theorem** must survive both proof review and theorem-level predecessor comparison, together with a nonroutine consequence. Examples of acceptable outcomes, not promises:

- a parameter-uniform principal-hull preservation classification, with the precise hypotheses and a proof mechanism not already a one-line corollary of existing theory;
- a generator-growth obstruction/attainability theorem with an explicit structural family and sharp upper bound in the stated class;
- a complete generator-profile theorem for a meaningful restricted class of embedded two-generated modules, showing why layer/extension/form data matter, and a useful classification or certificate consequence.

Insufficient alone: a new q/t/n table, an isolated μ3 witness, a finite exhaustive q4 rerun, matrix-rank substitution, the standard Nakayama formula, the known Morita threshold, or fixing a manuscript proof. A positive or negative answer to RQ1 by a short known corollary may resolve an old question but still fail this paper's novelty/depth gate.

7. Expected contributions and examples — all prospective

1. A source-delimited structural theorem for the retained principal-input class.
2. A rigorous principality/growth criterion and sharpness/preservation result, if obtainable.
3. A transparent role for nilpotent layers, embedded generator relations and the twist, rather than dimension-only claims.
4. Explicit certificate-ready examples and counter-controls illustrating **proved** hypotheses. No new parameter tuple is filled in now.
5. Optionally, a justified algorithmic or quantum corollary; neither is required to rescue an otherwise weak algebraic result.

Planned example tables separate input μ, hull μ, field dimensions, Loewy/extension data, generator matrices and proof status. They are specified in 06, not populated with invented outcomes.

8. Major risks, alternatives and stopping rules

Risk	Diagnostic	Bounded alternative	Stop/re-scope trigger
Exact result already exists	Primary theorem comparison with NL03/NL10/NL11 and cited predecessors	Identify a different nonroutine structural question and return to author	No substantive distinction after comparison
Core A is too small/routine	Answer follows immediately from standard normal form	Consider approved restricted e>2 class	Do not inflate q/r merely to make another table
Wild representation complexity	Proposed normal form requires all arbitrary modules	Restrict to a clearly defined two-generated embedded class	No meaningful tractable theorem class
Input invariants insufficient	Same proposed invariant cannot decide behavior	Retain additional pairing/extension/orientation data	Do not assert completeness of a coarse profile
PC1 is false	Exact witness verified independently	Seek a growth criterion and sharp family	An isolated computational counterexample is not a Q1 promise
PC1 is true but shallow	General theorem already forces it	Report as an old-question resolution; reassess paper aim	Do not repackage standard theory as principal novelty
Quantum bridge fails or is old	No metric/orthogonal proof or only old EA formulas	Publish algebra-only if strong enough	Drop quantum claims, not validity conditions
Old source/submission conflicts	Author supplies a different baseline/status	Re-inventory and delimit overlap before new project	No implicit reuse or correction of an unknown submitted version

9. What this document does not do

It does not prove PC1 or a new theorem, create examples, implement algorithms, edit MS03, select a journal, certify a quartile, establish priority or approve Phase1. The recommended research problem remains contingent on NAD01/NAD18 and the first-phase novelty gate.

BLUEPRINT DEVELOPMENT ONLY — WAIT FOR EXPLICIT AUTHOR APPROVAL.

04 — Proposed mathematical framework and assumption contract

SOURCE / 04_MATHEMATICAL_FRAMEWORK.md

BLUEPRINT ONLY: definitions and inherited facts, not new proofs.

Problem · Theorem roadmap · Quantum boundary

1. Notation, fixed before research

The old sources use r for the field degree in places. This proposal uses **m for field degree and r for matrix size**; translators must not identify the two.

Symbol	Meaning / assumption
p, q=p^m	p prime; m≥1 in background, m≥2 for a nontrivial coefficient-Frobenius programme
A=A_t=F_q[u]/(u^t)	Commutative finite chain ring, t≥1; coefficient automorphisms below fix u
R=M_r(A)	Matrix ring; **r=2 in the recommended primary programme**
X	**Central** indeterminate; no Ore/skew relation in P2-S
B=A[X]/(X^n−1), S=R[X]/(X^n−1)	B commutative; S≅M_r(B); cyclic codes are **right ideals** of S
σ_s	a↦a^(p^s) on F_q, extended coefficientwise fixing u,X; s modulo m
θ_s=σ_s∘transpose	Anti-automorphism; not assumed involutive
rev	X↦X^(−1) on B/S; coefficient reversal modulo n; automorphism, not anti-automorphism
Λ	Matrix trace followed by coefficient of u^(t−1), valued in F_q
B_s(x,y)=Σ_i θ_s(x_i)y_i	R-valued form, first slot twisted; y lies in the right orthogonal
b_s=Λ∘B_s	F_q-valued scalar form; σ_s-semilinear in first slot, linear in second
C^(⊥s)	{y:B_s(x,y)=0 for every x∈C}; not an unqualified ordinary Euclidean dual
Hull_s(C)	C∩C^(⊥s), always an intersection/submodule of C
μ_T(M)	Minimal number of T-module generators; μ(0)=0; not vector-space dimension
dim_q	F_q-dimension unless the field is explicitly changed

Recommended Core A: p=2, q=2^m, m≥2, t=2, n=2, r=2, all s. These are **proposed scope choices**, not approved assumptions for research execution. Wider t/e/r and a skew variable require separate gates.

2. Inherited results — inputs, not proposed contributions

The master mathematical blueprint and independent pairing audit control applicability. The old printed manuscript is not a blanket proof authority.

ID	Background statement retained under its exact hypotheses	Status and limitation
E01	A and R are finite Frobenius alphabets; the top-coefficient trace functional supplies the relevant nondegeneracy/ideal-catching structure	Established/imported background; do not infer same-sided perfect duality for an arbitrary form from nondegeneracy alone
E02	For right R-modules and the specified first-slot θ_s pairing, ring-valued and scalar orthogonals agree; duals have the size/dimension complement	Audited conditional mathematics; the manuscript's defective Hom/surjectivity proof is not imported
E03	Identical-twist double dual is **+σ_(2s)(C)**; universal reflexivity requires m divides2s (nontrivial matrix setting)	Inherited; codewise σ_(2s)-invariance is a different hypothesis; slot changes require translation
E04	Under the column Morita dictionary, W≤B^r corresponds to I_W={M:im_B(M)⊆W}; dim_q(I_W)=r dim_q(W)	Standard/inherited; columns for right ideals, not rows
E05	μ_S(I_W)=ceil(μ_B(W)/r); in particular I_W principal iff μ_B(W)≤r	Inherited packing threshold, **not** a new all-r theorem of this project
E06	For C=Σ_i g_iS, W(C)=Σ_i im_B(g_i) and W(C^(⊥s))=∩_i ker_B(θ_s(rev g_i))	Inherited, with right annihilator/reversal and sides fixed; no one-generator shortcut for a genuinely nonprincipal input
E07	W(Hull_s(C))=W(C)∩W(C^(⊥s))	Inherited hull reduction; for principal C=gS use im(g)∩ker(θ_s(rev g))
E08	CRT decomposes B into local repeated-root factors; μ_B(W)=max_j μ_(B_j)(W_j)	Standard/inherited; product of chain rings is not itself necessarily a local chain ring
E09	At r=2, if t=1 or p∤n, all component submodules need at most2 generators, so every cyclic right ideal, including every hull, is principal	Inherited generation result, **not freeness** of every submodule
E10	Coefficient expansion φ:C⊆R^n→F_q^N, N=tr²n, transports b_s to σ_s(x)J y^T with J=I_(r²n)⊗H_t in the fixed ordering	Inherited exact form/coordinate claim; not ordinary Euclidean/Hamming/CSS transfer

No proofs of E01–E10 are supplied here. Before use in a new paper, a future applicability audit must identify valid supporting arguments and retain their hypotheses. If that requires repairing MS03, that is a **separate old-paper correction task**, not silently authorized by a new research phase.

3. Proposed objects for the new question

For C=I_W, define

- **D_s(W):=W(C^(⊥s))**, using the actual global dual and the fixed Morita dictionary;
- **H_s(W):=W∩D_s(W)**;
- **generator-growth statistic Δ_s(W):=μ_B(H_s(W))−μ_B(W)**;
- **principality margin P_s(W):=μ_B(H_s(W))−r**;
- **principal-input spectrum**: the attainable profiles of H_s(W) among W≤B^r with μ_B(W)≤r.

These are **definitions/research bookkeeping**, not claimed new invariants or theorems. Their usefulness and external originality are unverified. Δ can be recorded without assuming its sign. A positive Δ does not alone mean a nonprincipal hull; that requires P_s(W)>0. A nonprincipal **dual** of a principal input, already known internally, is not the same as a nonprincipal **hull**.

A useful profile may retain: field length, μ, radical-layer dimensions, socle dimension/type, relations among minimal generators, off-diagonal extension parameters, and the restricted pairing/embedding

data. This list is **not asserted complete**. Equal abstract module type or equal field dimension need not determine how a module meets its dual; any claimed sufficiency is a future theorem obligation.

4. Local factors and their forms

Write n=p^a n_0 with gcd(n_0,p)=1 and factor X^(n_0)−1 over F_q. CRT gives local factors B_j; a coordinate model has the form

T_j≅K_j[u,w]/(u^t,w^e), K_j≅F_(q^(d_j)), e=p^a.

The exact coefficient-field embedding/isomorphism must be recorded, not assumed canonical. In a non-chain factor (t,e≥2), the maximal ideal is m_j=(u,w). **Nakayama's background identity** is μ_(T_j)(M)=dim_(K_j)(M/m_jM). Computing this quotient on a fully computed hull is not the proposed new theorem.

Core A has a single factor T=F_q[u,w]/(u²,w²) in characteristic2. However, its **duality is induced from the original coefficient pairing and reversal**, not chosen afresh as a convenient dot product on T². Reversal may act nontrivially on nilpotent coordinates in larger multiplicity; a linearized sign replacement is not licensed. For multiple factors, σ_s and reversal can couple different components.

A future local-form contract must specify:

1. source/target factors for every σ_s/rev map;
2. semilinearity over the correct coefficient/residue fields;
3. the induced functional/pairing matrix in the chosen bases;
4. which generator columns and kernels live in each module;
5. how global H_s(W) is reconstructed.

The known split, squarefree permuted-intersection formula is prior work. Establishing or verifying a more general translator is infrastructure unless it has independently demonstrated new content. No local orthogonal or reciprocal-factor formula is proved here.

5. Equivalence and normal-form discipline

- A change of generators of **the same** W preserves the input module/code. Exact canonical F_q-basis keys can identify equal subspaces after the B-action closure is known.
- Arbitrary left GL_r(B) changes the **embedded** module and generally its pairing relation. It is not automatically an isometry or hull-preserving equivalence.
- If a normal form uses an ambient change of basis, the pairing/adjunction and all structure operators must be transported too; discard neither the form nor the embedding parameters.
- A B-module isomorphism, a ring-code isometry, a coefficient-Hamming isometry and a physical Clifford/symplectic equivalence are different assertions.
- type(M)=dim_K soc(M) and μ(M)=dim_K(M/mM) must not be equated outside justified chain-ring situations; NL11 makes this distinction essential.

Thus the proposed classification is of a **specified embedded principal-input class with its induced form**, not all wild modules up to abstract isomorphism.

6. Correct dimensions, coordinates and metrics

For r=2, a Morita vector has2tn F_q coordinates; a word of R^n has4tn. If dim_q W=d, then dim_q C=2d. A field basis of the **hull Morita module** is not a full field basis of the image code. Feeding full image rows into a Morita unpair routine recreates MC11.

Coefficient expansion has $N=tr^2n$ and quasi-cyclic index tr^2 in the stated ordering, not index r. Its metric is coefficient Hamming, not ring-symbol Hamming or normalized homogeneous weight. A nonmonomial coordinate congruence may change minimum distance and requires recertification. In characteristic2, the alternating **bilinear** obstruction at even t does not forbid every Hermitian congruence or every code-specific CSS inclusion.

7. Explicit exclusions

No unqualified freeness, arbitrary automorphisms moving u, automatic Euclidean/trace orthogonal substitution, unverified skew quotient, generic gcd impossibility claim, conjectural DD-reflexivity assumption, or μ-to-entanglement identification is allowed. P3 would need its own Ore quotient and sided compatibility contract before adopting any of this central-variable framework.

No new theorem is proved or computational method implemented in this framework. All future work is approval-gated.

BLUEPRINT DEVELOPMENT ONLY — wait for explicit author approval before Phase1.

05 — Prospective theorem roadmap and dependency gates

SOURCE / 05_THEOREM_ROADMAP.md

PROPOSAL ONLY. No target below is proved, implemented or approved for execution.

Problem and PC1 · Framework · Phases

1. Status vocabulary and inherited boundary

- **E:** established/imported or inherited audited mathematics, conditional on exact hypotheses.
- **U:** unresolved inherited result/applicability question.
- **PC:** explicitly proposed conjecture, not a theorem or assumption.
- **RQ / TARGET:** a desired new question/result, not yet known to hold.
- **X:** externally unverified originality/priority or an incomplete comparison.
- **FINITE:** existing computational evidence, bounded by actual objects/range; not a universal theorem.

E01–E10 in 04 are prerequisites. In particular the Morita hull formula, μ-to-principality threshold, CRT/max-μ identity, chain-regime generation, double twist and Gray transport are **not** NT03–NT09's new results. The old generic μ3 direct-sum proof fails; valid separate witnesses do not repair it. Old H-3A.1/H-3A.2 are not used as premises.

PC1 is the exploratory Core A principal-hull-preservation conjecture in 03; Arena recommends question-only posture until literature/applicability review. No target depends on PC1 being true.

2. Target cards

NT01 — Source-valid inherited input contract

Type/status: E-applicability work, NOT a novelty claim; pending.

Scope: r=2, first-slot θ_s/right ideals/central X, beginning t=2,n=2,p=2. Identify exact valid supporting results for E01–E10 and document manuscript proof defects that must not be inherited.

Required output after authorization: a lemma-use ledger with source/proof locator, quantifiers, side, slot, scalar field, and unresolved cells. If a valid premise is unavailable, stop or explicitly request separate proof/correction authority. This is not permission to edit the old paper.

Exit: no core dependency relies on a known invalid proof, wrong field encoding, stale “open” status or an unreviewed side/form substitution. The claim count is not a proof count.

NT02 — Faithful local-form and embedding contract

Type/status: infrastructure TARGET; originality X, normally background rather than a main theorem.

Desired statement: on the retained local model, exhibit an exact dictionary between global principal matrix cyclic inputs, their embedded two-generated Morita modules, and the **induced** local dual/hull condition. For any approved multi-factor extension, identify source/target factors and their twist/reversal transport rather than assume independent local hulls.

Dependencies: NT01 and standard CRT; no assumed standard local dot-product form. Existing split/permuted formulas must be acknowledged.

Proof obligations later: well-defined domains/codomains, both directions of the dictionary, scalar-field semilinearity, independence from generator presentation, and preservation of the actual hull submodule. Verification of a familiar dictionary is not automatically new mathematics.

Exit: a mathematically coherent model against which NT03 can be stated; no full-wild or quantum theorem inferred.

NT03 — Core generator-profile structure theorem

Type/status: principal research TARGET / RQ2–RQ4; truth and external novelty unresolved.

Desired result: a structural normal form or finite set of explicitly specified invariants for the **retained embedded principal-input class** that determines μ_T(H_s(W)) and relevant hull layers without enumerating all ambient modules. Begin Core A; extend only if a nonroutine theorem is possible.

What must be discovered, not assumed: which minimal-generator relations, radical layers, extension parameters and restricted form/embedding data are sufficient. A basis-independent statement must account for allowed changes of generators and transported ambient forms.

Proof obligations later: existence and coverage of the proposed form/class, reconstruction, exact sufficiency of the data, no loss of pairing information, correct zero/boundary cases, and parameter/twist quantifiers. If uniqueness is claimed, it needs its own proof.

Novelty failure test: writing μ(H)=dim(H/mH), recomputing a field nullspace, or using an existing MT GPM followed by standard linear algebra is insufficient unless a genuinely additional structural theorem is demonstrated.

Exit: a full proof in a future new-research workspace and independent theorem-level comparison with MS03, NL02/NL03/NL10/NL11 and relevant cited algebra. No proof is supplied now.

NT04 — Principal-input dichotomy / principality criterion

Type/status: principal research TARGET / RQ1,RQ3; direction open.

Desired result: characterize precisely when μ_T(H_s(W))≤2 for the retained two-generated inputs. Two legitimate branches are:

- **Preservation branch:** establish preservation for a parameter-uniform class, with exact hypotheses and an explanatory reason; assess whether it is nonroutine enough for the paper.
- **Growth branch:** specify an obstruction/criterion yielding μ>2, with a structural family and independently checkable explicit representatives.

Dependencies: NT02; preferably NT03 for an explanatory criterion. PC1 is not a dependency. A theorem valid only at q4 by finite census is not the intended universal contribution.

Exit: criterion is necessary/sufficient in the stated class or is honestly labelled only one-sided; all quantified claims proved. A counterexample must start from **μ(W)≤2** and have μ(H)>2. A nonprincipal dual or input is not a substitute.

NT05 — Sharp bounds and attainability within the retained class

Type/status: TARGET / RQ5; conditional on a useful NT03/NT04.

Desired result: a sharp upper bound, attainable generator spectrum, or preservation classification that explains what hull generator numbers are possible under the selected input/form constraints.

Proof obligations later: upper bound over the entire stated class, attainment of every claimed extremum/value or explicit exclusions, and separation of field length, μ, type and matrix-ideal generator number. If no growth occurs, do not manufacture a positive example; characterize the preservation boundary instead.

Novelty test: ordinary submodule bounds, standard socle/type dualities and the known matrix packing threshold are compared before any new-bound label. A short specialization of NL11 is not an independent contribution.

Exit: sharpness claimed only where supported by a general proof and explicit certificate-ready examples, not a search maximum.

NT06 — Twist and embedding dependence

Type/status: TARGET / RQ4,RQ6; optional co-main result if substantial.

Desired result: determine how changing s, or changing embedding data at fixed coarse module type, changes the criterion/profile in the approved class. Include non-reflexive m∤2s only when the exact local/global dual condition is retained.

Dependencies: NT02–NT04. Repeating DD=+σ_(2s) or observing hull-dimension variation is prior work, not this result.

Exit: any sufficiency/invariance statement has a proof; counterfamilies distinguish genuinely relevant data from a mere coordinate re-encoding. If no new dependence survives, record that rather than invent a twist advantage.

NT07 — Global synthesis under coupled CRT factors

Type/status: conditional extension TARGET / RQ7, not required to start.

Desired result: determine whether the local principal-input criterion assembles under actual factor coupling, yielding a meaningful global classification or new obstruction. The component permutation, coefficient-field maps and induced forms must be explicit.

Dependencies: NT02–NT05 and independent review of any new generalization. Standard CRT and μ=max(component μ) are background.

Exit: nonroutine content beyond reassembling already known component formulas; exact hypotheses on multiplicity, factors and twists. If it is only a routine corollary, identify it as such, not a second main theorem.

NT08 — Certificate-producing structural computation

Type/status: conditional algorithmic TARGET / RQ8; not implemented.

Desired result: an algorithm on the retained normal-form/invariant input that outputs the hull generator profile and, when claimed, explicit minimal generators and principality certificates, with a justified cost bound in the stated input model.

Dependencies: NT03–NT05 and an independently specified arithmetic/coordinate contract. Proposed implementation/testing is in 06.

Proof obligations later: termination on the selected class, correctness, completeness of any enumeration, minimality certificates and a reproducible complexity statement. A finite agreement count cannot prove the algorithm's universal correctness.

Exit: proved algorithm claim plus independent checks on named objects; if only a useful reference implementation is achieved, label it supporting computation rather than a theorem-level contribution.

NT09 — Optional quantum corollary

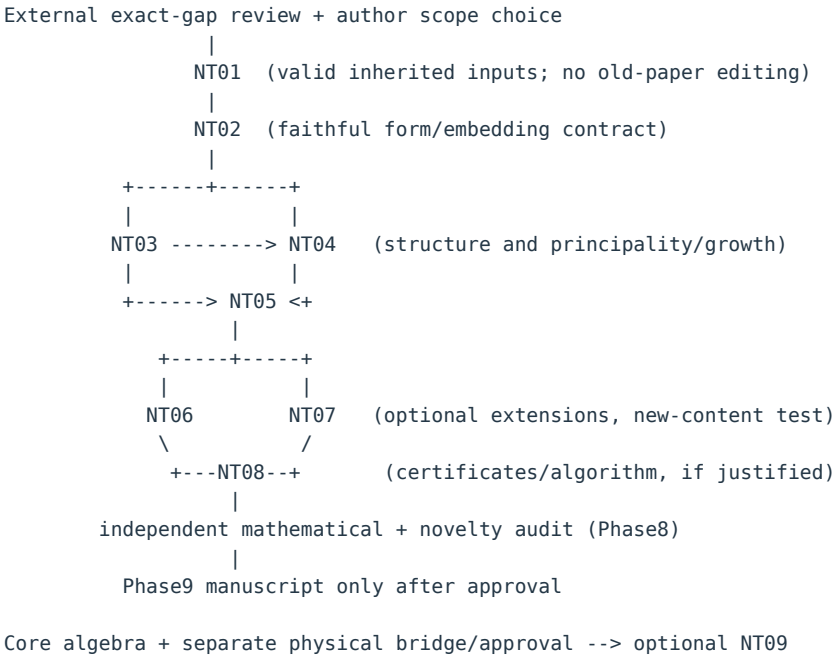
Type/status: optional RQ9 / TARGET, **no bridge established**.

Desired result: only if possible, a new structural consequence for an actual Euclidean/Hermitian/trace hull or commutator radical, with a justified metric/physical-label map. Standard EA/CSS formulas are imports; μ(H_s) is not entanglement cost.

Dependencies: core algebraic results, all gates in 07, separate author authority and new-paper audit. Failure does not force abandonment of a sufficiently strong algebraic paper.

Exit: complete physical hypotheses, field/trace dimensions, phase/lift or Bob completion, route-correct logical-error set, distance status and matched-resource comparisons. If absent, omit quantum claims rather than weaken these conditions.

3. Dependency structure



NT03 and NT04 may be reorganized after an authorized pilot; the graph is not an execution order granted by this document. Neither an old correction dependency nor a passed computation closes a new theorem gate.

4. Minimum contribution and claim-promotion rules

Before a new-paper GO recommendation: at least **one nonroutine structural main theorem** plus a meaningful proved consequence/classification must survive independent review and external comparison. Multiple routine lemmas do not meet this by count. Any target may end as: **proved under stated restrictions / refuted with a verified witness / computational evidence only / unresolved / already known / withdrawn**.

For every promoted claim require an exact statement, complete proof locator, reviewed dependencies, explicit range, separate computational evidence, and separate novelty evidence. “All tests PASS” cannot promote TARGET, PC or X to theorem or priority. If only routine translations or old-paper repairs remain, the correct outcome is **STOP/RE-SCOPE**, not a forced Q1 claim.

BLUEPRINT DEVELOPMENT ONLY — no proofs or research phase started.

06 — Computational methodology, example design and certificate plan

SOURCE / 06_COMPUTATIONAL_METHODOLOGY.md

DESIGN ONLY. No scientific implementation, import, experiment, enumeration or new code construction is performed in this task.

Framework · Theorem targets · Future phase gates

The scripts in this delivery copy/hash/check/package planning files only. They are **not** prototypes of the following research programme.

1. Purpose and separation of evidence

Future computation, if expressly authorized, should falsify overly broad structural claims, verify explicit objects and support proofs. It must not substitute a finite scan for a parameter-uniform theorem, establish priority by no-match searches or claim optimal quantum parameters by sample maxima.

Separate labels for:

1. original-source execution and its side effects;
2. equality of the **actual** input object after coordinate/field translation;
3. independent arithmetic on that object;
4. completeness of an input-family enumeration;
5. exactness or lower-bound status of a distance;
6. universal proof status and external novelty, both separate from1–5.

No old campaign is to be rerun merely to fill the new paper's tables. The completed28/51-code audits and1,221-module census are reusable **reference evidence**, not newly generated results.

2. Future arithmetic and representation contract

Layer	Required contract before implementation
Finite field	p,m, explicit irreducible polynomial, ordered polynomial basis, integer encoding, arithmetic version and field-isomorphism maps where needed
A_t and B	u/X coefficient order, truncation, reduction modulo X^n−1, cyclic shift; all units/zero tests unambiguous
Local T_j	Explicit global-to-local isomorphism and inverse, K_j embedding, u/w ordering, induced σ_s/rev/form operators
Matrix ideal	Right ideals and **column** Morita modules; r=2 fixed unless separately approved
Morita basis	Canonical F_q-basis in F_q^(2tn) for W and H_s(W), not merely a B-generating list
Image-code basis	Separate F_q-basis in F_q^(4tn), with explicit maps to/from Morita representation; dimensions checked
Equivalence	Exact subspace identity keys distinguished from generator changes, ambient GL or physical isometries
Duals/hulls	Actual first-slot Galois form, ring/scalar comparison where valid, induced local form; Euclidean/trace outputs named separately

For reuse of old matrices, the actual encodings are GF8: **x^3+x+1**, GF9: **x^2+1**, GF4: x^2+x+1. Earlier alternative library polynomials do not define identical integer labels. New work may use another encoding only with an explicit checked isomorphism and provenance; it must not rewrite inherited bytes.

3. Independent calculation paths — future design

- **Path A (definition):** compute the dual using the specified scalar/ring pairing and field linear algebra; intersect with the input code, then convert to its Morita module under a validated representation map.
- **Path B (module structure):** use the source-reviewed column image/kernel/reversal description and the proposed structural invariants/normal form; construct the hull module directly.
- **Path C (small exact control):** on explicitly tiny cases only, enumerate module elements or candidate generators to validate closure, equality and minimal-generation certificates.

A and B must compare **subspaces**, not only dimensions. If both use the same flawed field routine or the same hull helper, agreement is not independent arithmetic. A later verifier should use separately implemented arithmetic/linear algebra or a justified external system with frozen versions. Shared standard mathematics is acceptable; shared unexamined implementation is disclosed.

Minimality is certified by the residue-space dimension H/mH over the correct K and a generator lift whose module closure equals H. Not finding a small generating set in a random search is not a minimality proof. Generic Nakayama arithmetic is a **check of an object**, not by itself a new structural theorem.

4. Planned input families and example roles

No row below reports a new outcome.

Planned family	Purpose	Proposed mode / limit	Expected evidence, not expected answer
Existing (4,2,2), s=0,1, r=2	Regression/old finite anchor; principal-input filter	**Reuse sealed existing census**, not rerun the campaign in this planning task	Known1+3 nonprincipal hulls belong to nonprincipal inputs; old data only
q=2, t=2,n=2	Tiny arithmetic/negative controls if approved	Exhaust only if the chosen object count is proved small	No nontrivial Galois evidence at a prime field
q=8, t=2,n=2, s=0,1,2	Non-reflexive comparison in proposed Core A	Normal-form-directed finite class or capped sample, never a claimed full ambient census	Explicit profile/counterexample or no finding **within the actual range**
Symbolic q=2^m Core A	Test structural theorem hypotheses and parameter dependence	Future symbolic reasoning separately authorized; not replaced by many numerical q	Parameter-uniform theorem only with proof
t=2,e=3,p=3 local factor, e.g. q9,n3	Odd-characteristic/repeated-root contrast, only after Core A gate	Small chosen principal-input classes/capped examples	Exact induced form and named profiles, not a prefilled μ3 growth claim
Several CRT factors	Validate coupling and any genuine NT07 extension	Only after approved local result and explicit transport	A nontrivial coupled example if one exists; naive independent-hull control
r=3 or larger t/e	Optional later extension	**Not part of the initial pilot**	Requires new structural justification, budgets and approval

Expected worked examples: zero and ambient codes; principal LCD and principal self-orthogonal inputs; an example distinguishing embedding/form data if it exists; preservation/growth boundary examples dictated by a proved theorem; a nonprincipal-input old witness clearly labelled **prior work**. No example category is guaranteed nonempty.

5. Planned table schemas — no fabricated parameter rows

Table A — algebraic generator profiles

object_id; origin (old/new); q/p/m; field modulus; r/t/n; local factors; s; input generators and hash; dim_q W; μ_B(W); input principal certificate; dim_q H; μ_B(H); local radical/Loewy profile; relevant embedding/form data; P_s; proof/theorem locator; verification mode; exactness/completeness; limitations.

Table B — principal-input spectrum/classification

class_id; exact class definition; number expected/proved if available; number visited; completeness argument; twists; attained hull μ values; bound; attainment witness IDs; unvisited/timeout count; theorem versus finite status.

Table C — algorithm/certificate costs

`input model; D; coefficient field; arithmetic operation count if proved; wall time/hardware; memory; candidate count; rejected/accepted/duplicate counts; verifier time; version/seed/budget; timeout or incomplete status`.

Table D — optional quantum parameters

Only after the separate quantum gate: actual field-code matrix/hash; physical q; classical construction alphabet; N; k; h_E or exact trace-radical dimension; route; K; c and resource convention; d exact or ≥L; certificate method/range; Bob/phase check; source of external comparator and returned physical header; novelty qualification.

An algebraic μ column is never substituted for an h_E column. Repeated parameter tuples, route-labelled rows, equal matrices and equivalent codes are counted separately and named explicitly.

6. Certificate bundle per named new object — future requirement

At minimum retain:

1. exact field/ring/coordinate metadata and generator bytes;
2. input and hull canonical field bases with separate module/image dimensions;
3. B-action closure witnesses and right-sided generator reconstruction;
4. dual/hull bases from both paths, or an explicit unresolved disagreement;
5. local H/mH bases, radical products and generator lifts;
6. rank/nullity checks, input μ≤2 certificate and hull μ value;
7. theorem-hypothesis flags, including t/e and reflexivity distinctions;
8. independent verifier output, source/program hashes and execution environment;
9. enumeration scope or sample seed/selection rule, rejected/duplicate/timeout counts;
10. if quantum retained, all physical checks and distance evidence separately.

An object with matching parameters but different matrices does not inherit another certificate. Any coordinate transformation or field migration invalidates byte-level reuse until its map and any weight change are checked.

7. Verification and falsification tests to specify before running

Future tests must include: wrong left/right action; rows-versus-columns; omission of rev at **n≥3** (n≤2 is vacuous); wrong twist orientation; local factor permutation omitted; illegal freeness assertion; coarse-dimension-only classification; full image rows passed as Morita vectors; bilinear/Hermitian form confusion; ordinary-Euclidean versus transported hull substitution; incorrect GF8/GF9 labels; duplicate generator presentations counted as distinct codes.

If quantum is approved: include odd-characteristic phase sign, trace tests on a full Fp basis, Fq/trace orthogonal mismatch, missing Bob completion and SO/DC logical-set mismatch. Do not report a negative control as passed when the chosen test case makes it tautological.

8. Complexity and proposed resource envelope

Let D=2tn be the Morita field dimension at r=2; full image dimension is2D. A local T_j^2 has K_j-dimension2te. Dense linear algebra has ordinary polynomial-size costs in the expanded dimension, but **enumerating subspaces or generator tuples is exponential**; a precise algorithm bound is NT08, not asserted here.

Naively scanning every pair of generator columns already involves q^(2D) presentations before deduplication. Enumerating all submodules in F_q^D can be vastly larger than listing the few relevant normal forms. Full-ring or rank-three exhaustive search is not the fallback.

Proposed initial pilot cap, NOT AUTHORIZED: at most3 explicitly listed parameter families; at most2,000 distinct candidate inputs per new family unless a proved complete smaller class is chosen;

total2 CPU-hours,4GiB RAM,250MiB new artifact budget. The author may choose smaller/larger limits or no computation (NAD10). Preflight must count/upper-bound candidates and estimate memory; stop at budget rather than silently switch to sampling and keep an EXH label. Large later data stay outside Git according to an author-approved storage/manifest policy.

These are safeguards, not performance promises. No timer, sample or resource usage from a new scientific run is reported now.

9. Completeness and distance vocabulary

Use **EXACT_OBJECT_LINEAR_ALGEBRA**, **EXHAUSTIVE_STATED_FAMILY**, **EXHAUSTIVE_PAIRS**, **SAMPLED**, **PARTIAL/TIMEOUT**, **DISTANCE_EXACT**, **DISTANCE_LOWER_BOUND**, and **NOT_COMPUTED** as separate fields. Enumerating all ambient words does not imply testing all word pairs. Exhaustive invariant enumeration does not imply all distances were evaluated.

For any later distance: complete enumeration or a complete increasing-support search through an attained minimum may certify exactness; excluding weights≤L without finding a witness gives d≥L+1. An ambient search lower bound is not an exact independent distance. The historical M1=24/M2 fallback=4/M3 exact11+LB17 counts are not a template for declaring “all three methods” on new records.

10. Safe future execution boundary

Scientific work, if later approved, uses a fresh disposable successor workspace, `python3 -B`/`PYTHONDONTWRITEBYTECODE=1`, immutable inputs and side-effect paths outside sealed evidence. A relocated historical extractor needs its recorded `MATRIX_RING_COMP`; do not run it again gratuitously. New implementations should not write beside old research data. No existing scientific script is executed or modified by this blueprint.

Exit for computation: every retained new object has a typed certificate and honest completeness label; disagreements/timeouts remain visible. **A computational PASS is not a theorem, novelty, quantum-validity or Q1 PASS.**

BLUEPRINT DEVELOPMENT ONLY — wait for explicit author approval before Phase1.

07 — Optional quantum strategy: physical hypotheses before applications

SOURCE / 07_QUANTUM_CONSTRUCTION_STRATEGY.md

Default recommendation: an algebra-first paper. Quantum applications are optional and separately gated. No new quantum code, bridge or parameter is claimed.

Theorem roadmap · Computational design · NAD07

1. Why quantum is not the primary novelty claim

P2-S asks about **minimal module generator number** of the actual Galois hull. Standard EA parameters depend on a specified physical commutator radical or the ordinary Euclidean/Hermitian hull of a field code. These are different invariants. There is no proved μ_B(H_s)→h_E or μ_B(H_s)→c map in this blueprint.

MS03 already has quantum constructions and four supplied q8 EA records; the earlier A8 matrix-ring Gray/CSS construction and field/commutative Galois-hull EA papers are mandatory predecessors. A new table made by the same generic EA formula would not make the new structural result a new quantum construction.

NAD07 recommendation: retain a conditional application track, but exclude quantum from the minimum contribution requirement. If no genuinely useful and justified connection survives, remove quantum claims rather than force them into the title.

2. Coordinate and physical-alphabet contract

For coefficient expansion of R=M_r(A_t), N=tr²n **field coordinates** over F_q. At r=2, N=4tn. The associated symplectic label space has2N field coordinates; that is not2N physical qudits.

The transported form is σ_s(x)J y^T, J=I_(r²n)⊗H_t in a fixed ordering. It is not automatically ordinary Euclidean. Involutive Hermitian, general sesquilinear, bilinear and absolute-trace forms must be distinguished. A field code over F_(q²) used in a Hermitian construction may correspond to physical q, not physical q²; every record must state both alphabets.

A nonmonomial congruence can change Hamming distance. The characteristic-two/even-t **bilinear** obstruction does not prohibit Hermitian congruences or all individual Euclidean self-orthogonal images. No ambient-form assertion replaces an actual code inclusion or physical isotropy test.

3. Standard routes available later — imports, not new results

Let D≤F_q^N be a **specified F_q-linear** field code of dimension k, and H_E=D∩D^(⊥E), h_E=dim_q H_E. The following single-code Euclidean routes are inherited standard constructions under their physical stabilizer hypotheses; no proof is developed here. Distance statements use nonempty logical-error sets, e.g. K>0; zero-dimensional codes require separate conventions.

Route	Required actual hypothesis / stabilizer input	Parameters	Relative-distance set
SO CSS	D⊆D^(⊥E), with phase-correct scalar-free lift	K=N−2k, c=0	D^(⊥E) minus D

Route	Required actual hypothesis / stabilizer input	Parameters	Relative-distance set
DC CSS	D^(⊥E)⊆D; stabilizer is built from D^(⊥E), **not from the original D as if SO**	K=2k−N, c=0	D minus D^(⊥E)
Euclidean EA	D arbitrary F_q-linear; valid Bob completion of the nonisotropic Alice labels	c=k−h_E, K=N−k−h_E	D^(⊥E) minus H_E

For EA, c denotes shared maximally entangled **q-dimensional pairs** in this convention; do not silently equate it with a qubit-ebit count. Bob completion and an Alice-only logical-error argument are required; unextended noncommuting Alice labels do not all fix a codespace.

The distance is the minimum **coset-leader** weight of the route-correct relative set, not the Hamming weight of an arbitrary quotient representative. A quotient does not automatically become an ordinary length-N Hamming linear code.

The EA identities K−c=N−2k and K+c=N−2h_E are not universal formulas for R1 or DC with original k. The audit's742 DC metadata failures do not refute their code distances; they refute the blanket bookkeeping scope. No such universal statement will be reused.

4. General non-reflexive/trace route — only if explicitly justified

A proposed twisted graph route must identify its exact label space L and linearity field. A map involving coefficient Frobenius can make L **only F_p-linear**, even when its source D is F_q-linear. The commutator is the **absolute trace-symplectic** form. Its trace orthogonal generally differs from the orthogonal defined by an F_q-valued symplectic equation.

For an F_p-linear isotropic label space of dimension a, a scalar-free phase-correct stabilizer lift would have codespace dimension p^(mN−a). Writing that as q^K requires a=m(N−K) with the appropriate integer K; an Fq-linearity assertion cannot be inserted to force this. A graph construction with a=mk is a special case to justify, not an assumption about every label space.

Future proof/verification obligations:

- exact ordered X/Z/Weyl convention and odd-p phase sign;
- trace isotropy on an Fp basis, or genuinely equivalent full-value conditions;
- a scalar-free group/lift with correct p-dependent phases;
- trace centralizer and actual stabilizer subgroup intersection;
- Bob resource accounting if labels are nonisotropic;
- correct physical weight/error model and a nonempty logical-error set.

The existing R1 is prior work, not NT09's novelty. A “semilinear entanglement number” requires a new, valid physical derivation and predecessor comparison, not replacement of h_E by h_s in a formula.

5. Gate for connecting the new algebra to quantum information

A quantum application may proceed only if **all** of QG1–QG6 are satisfied in a later approved phase:

Gate	Evidence required
QG1 — algebraic relevance	A proved new structural theorem identifies the actual invariant controlling the selected physical construction, or a clearly limited new consequence

Gate	Evidence required
QG2 — form bridge	A proved map to the relevant Euclidean/Hermitian/trace form, with fields, sides and domains explicit; or direct physical commutator calculation on each actual field-code matrix
QG3 — metric bridge	Monomial/isometric justification where claimed; otherwise recomputed distances in the physical coordinates, no Hamming-preservation assumption
QG4 — physical validity	Isotropy/phase/scalar-free lift or Bob completion and route-correct orthogonals, with an independently reviewed proof
QG5 — named-object certification	Exact matrices, field encodings, k/h/K/c and inclusions checked; exact d or honest lower bound; no tuple-only certificate reuse
QG6 — genuine incremental content	Source-specific comparison with MS03, A8, Cao, C5/C6, NL08 and other relevant works; not a generic EA formula plus another table

Direct ordinary-Euclidean verification can justify a **particular application** without an ambient Galois→Euclidean theorem. It does not turn a Galois μ-classification into an entanglement-cost theorem. If only such generic applications are available, label them illustrations or omit them.

6. Prospective examples and benchmark policy

After QG1–QG6 and author approval, examples could pair an algebraic principal-input classification certificate with a fully specified field image and a correct SO/DC/EA/trace route. No new tuple is predicted now. In particular, neither principal input nor μ3 hull implies distance improvement or low entanglement.

Compare only identical physical alphabet, N,K, distance status, c/resource convention and route/bound hypotheses. Record the returned table header and bound interval, not just a query parameter. Physical-q8 stabilizer tables and qubit/qutrit EA tables exist at audited probes; absence of a guessed endpoint does not establish absence of all tables. Those tables do not automatically benchmark physical q4/q8/q9 EA examples.

No “record”, “optimal”, “first q8”, universal budget ceiling or new Singleton-bound priority is claimed. B047's inherited $d \geq 5$ at $c=2$ already contradicts the old $d \leq 4$ low-budget ceiling; it is not a new construction of this proposal. A family sample can contain an exact-distance code, and a full invariant census can still have a capped distance stage.

7. Quantum no-go/omit option

If the new structural invariant has no nonroutine physical consequence, if the bridge is already known, or if the metric/trace hypotheses cannot be proved within the authorized scope, end NT09 as **UNRESOLVED/ALREADY KNOWN/OMITTED**. A strong algebraic result can proceed to its own journal-fit assessment without quantum. A weak algebraic result cannot be rescued by adding invalid quantum claims.

BLUEPRINT DEVELOPMENT ONLY — no quantum construction or implementation authorized.

08 — Phase-wise execution plan, measurable gates and journal positioning

SOURCE / 08_PHASE_WISE_EXECUTION_PLAN.md

CURRENT STATE: Phase0 planning only. Every future phase requires explicit author approval.

Master · Decision register · Theorem roadmap

1. Governance and separation from the old paper

The old audit remains **CONDITIONAL HOLD** with its independent wording: **CONSOLIDATED AUDIT ONLY — IMPLEMENTATION NOT AUTHORIZED. Wait for author approval.** None of its43 correction families or20 author choices is closed by this new programme. A future new-research phase does not authorize edits to MS03, historical scripts, data or completed seals.

The first future phase proposed here is a **literature/source/scope gate**, not proof development or experiments. Approval of it is not approval of Phases2–10. Completion of any phase triggers a report and a **STOP for the next explicit decision**, not automatic progression. If work exceeds the chosen class, theorem or budget, request reauthorization before proceeding.

2. Complete roadmap at a glance

Time ranges are rough effort estimates **after approval**, not scheduled dates, promises of success or permission to work.

Phase	Objective	Prospective effort	Mandatory exit / decision
0 — current	Develop and seal this new blueprint, preserving sources	Current planning task	Ten documents,18-topic map, pending decisions, integrity checks; stop
1	Exact-gap, source-authority and scope review	1–2 weeks	Primary theorem comparison and author scope choices; GO / NARROW / STOP
2	Valid inherited-input and local-form specification	1–2 weeks	NT01/NT02 contracts, no unresolved load-bearing premise
3	Bounded computational falsification pilot, if approved	1 week within explicit cap	Typed reference implementation/certificates, honest sample/completeness and negative controls
4	Core structural theorem programme	3–6 weeks	Full proofs or explicit refutations/unresolved outcomes for NT03/NT04
5	Sharpness, twist and approved structural extensions	2–4 weeks	NT05 and justified NT06/NT07, no size-only extension
6	Independent certificates and algorithm review	1–2 weeks	NT08 if proved; every retained new object independently checked
7 — optional	Physically valid quantum consequence	1–3 weeks, separate authority	QG1–QG6 or omit; no automatic algebra→quantum transfer
8	Independent mathematical, numerical and novelty audit	1–2 weeks	Claim-by-claim closure, overlap review and conditional paper GO or STOP

Phase	Objective	Prospective effort	Mandatory exit / decision
9	New manuscript preparation only after Phase8 approval	2–3 weeks	New source/PDF/theorem/table traceability and author review
10	Final journal/publication gate	Variable	Verified journal/metric/policy facts, author approval; submission separately authorized

The full programme may take materially longer or terminate early. None of these estimates implies a Q1 result will exist.

3. Detailed work packages and exit criteria

Phase0 — blueprint delivery (authorized now)

Permitted: source/report/register study, bounded literature retrieval, comparison, proposal writing, file integrity tooling and ZIP packaging.
Forbidden: new proofs, scientific implementations/imports/experiments, new examples, old corrections, manuscript edits/builds, commits/pushes or editorial action.

Measurable exit: all ten requested narratives exist; all18 topics map to substantive sections; P1/P2/P3 compared with weighted scoring; every pending new author decision has alternatives, recommendation, mathematical justification and consequences; all43 audit families and16 RF findings remain linked; original preservation and ZIP/hash checks pass. These are **delivery criteria**, not mathematical GO.

Phase1 — first future phase: exact novelty and scope gate

Entry: NAD01 and NAD18 explicit approval; exact old-source/submission facts requested under NAD02; provisional class/twist boundaries agreed. If the author cannot yet identify a later source, that uncertainty remains visible and blocks affected overlap claims.

Work: inspect exact primary theorem statements and dependencies of MS03, foundational anti-isomorphism/form literature, NL01–NL06, NL10–NL11 and their relevant cited predecessors. Resolve the MT2025/2026 version relationship if relied on. Record access failures, not invented content. Search by related module/intersection/syzygy terminology, not only “matrix Galois hull”. No theorem proving or scientific implementation in this phase.

Required artifacts: exact-gap comparison ledger; primary-source/version/bibliography ledger; old/new contribution boundary; scoped question/target revision; unresolved-access/metadata register; author-decision update.

Measurable exit:

1. At least12 relevant theorem/source comparison rows, covering MS03, the decisive algebraic foundations and NL01/NL02/NL03/NL04/NL05/NL06/NL10/NL11; a count alone is insufficient.
2. Each proposed core claim NT03/NT04/NT05 has at least one direct primary-theorem comparison, or remains explicitly X/blocked.
3. No load-bearing positive gap conclusion depends solely on an abstract's silence.
4. Explain whether the candidate is a routine specialization of scalar hull classification, GPM/intersection generators or Artinian duality bounds.
5. Author confirms the research class, twist scope, acceptable negative outcomes and independent-review route.

Outcome: GO to a **separately approved Phase2**, NARROW and return for approval, or STOP if no substantive unoccupied problem remains. This is a reasoned bounded novelty assessment, not proof of all-literature absence.

Phase2 — foundations and faithful local specification

Entry: Phase1 review accepted and Phase2 explicitly authorized.
Work: NT01 inherited-lemma applicability ledger; NT02 local/global coordinate and form contract; precise principal-input class and allowed equivalences. Review existing proofs, without editing the old paper. If genuinely new proof work is needed beyond this specification, request a scope amendment rather than smuggle it into source review.

Measurable exit: every imported lemma lists side/slot/scalar field/quantifiers and valid proof location; all local maps have domains/codomains/inverses where claimed; input/hull representations and factor coupling are unambiguous; no freeness, automatic Hermitian, Euclidean or physical assumption remains hidden. The first pilot's finite cases and failure controls are specified before code.

Outcome: approve or reject the exact research contract; then stop for Phase3/4 authority.

Phase3 — bounded pilot and falsification (not currently authorized)

Entry: explicit approval of implementation and the selected cases/resource envelope under NAD10/NAD14.
Work: construct a **new** disposable research implementation according to 06, not a repair of old scripts. Reuse sealed old evidence as reference; compare named objects via actual matrices. Run a bounded pilot on approved classes, with independent arithmetic and negative controls.

Proposed cap: ≤3 families, ≤2,000 distinct new inputs/family unless a smaller complete class is proved, total2 CPU-hours/4GiB/250MiB artifacts. Not permission to spend these resources now.

Measurable exit: every visited object has field/coordinate metadata, closure, actual hull subspace and local μ certificate; two calculation paths agree as subspaces or flag discrepancies; every control's nonvacuous scope is recorded; sample/exhaustive/timeout/distance statuses distinct; no old input changed. A search without counterexamples leaves PC1 unproved.

Outcome: choose preservation, growth or re-scope as a **research strategy**, not a mathematical verdict; obtain separate theory approval.

Phase4 — core structural theory

Entry: approved exact NT03/NT04 statements/classes, informed but not proved by the pilot.
Work: seek complete structural and principality/growth proofs, allowing restriction, refutation or failure. No result is demanded to be positive.

Measurable exit: every claimed theorem has a complete proof and dependency list; the claimed class/normal form has a coverage argument; the criterion is explicitly necessary/sufficient or only one-sided; proposed invariants are not claimed sufficient without proof. A growth example has μ(input)≤2 and μ(hull)>2. A preservation theorem states parameter/twist restrictions. External comparison is updated at the **final** statement level.

Outcome: if only an old theorem or a routine corollary remains, stop/re-scope; no forced paper. Independent proof feedback precedes extension.

Phase5 — sharpness and structural extension

Entry: accepted core results and author-selected NT05/NT06/NT07 scope.
Work: determine a sharp spectrum/bound/preservation boundary, explain twist/embedding dependence, and only then consider e>2 or multiple factors. Matrix-rank3 is not an automatic next step.

Measurable exit: every extremum has an upper-bound proof and an attainment/exclusion argument in the retained class; global synthesis respects actual coupling; routine CRT and known max-μ facts are marked background. New examples have explicit source/matrix identity. Unsupported generalizations are withdrawn or remain open.

Phase6 — independent computation and algorithm certificates

Entry: proof and dataset scope frozen for review.
Work: independent implementation/verification of retained examples, generator minimality and NT08 if claimed; resource/time/completeness reporting. Distance work only for explicitly approved quantum objects.

Measurable exit:100% of retained new named examples have exact algebraic certificates; no unknown distance is printed as exact; every enumerated class has a completeness argument or a sample label; no source flags masquerade as independent certificates; tables and theorem hypotheses agree. A verifier must catch at least the predeclared wrong-side/form/representation negative controls where meaningful.

Phase7 — optional quantum bridge

Entry: NAD07 approval and a defensible proposed link to the core theorem.
Work: NT09 under 07; separate physical fields, trace groups, phase/scalar-free lift or Bob completion, relative distance and resource comparisons.

Measurable exit: QG1–QG6 all satisfied for every retained route/claim, with independently certified named objects. Otherwise mark the track omitted/unresolved. Algebraic paper assessment proceeds independently; quantum failure cannot be concealed as a Gray-map assumption.

Phase8 — independent new-paper audit (before drafting)

Entry: proposed final theorem/claim set, new example certificates and literature comparison frozen for review.
Work: independent mathematical and computational review; proof-statement consistency; predecessor/own-work overlap; type/field/form/metric scope; optional physical validity; no priority based on finite counters.

Measurable exit:

- At least one nonroutine structural main theorem plus a meaningful proved consequence/classification survives, judged for substance rather than lemma count.
- Every retained universal claim is independently reviewed with no unresolved critical dependency; others are restricted, withdrawn or explicitly open.
- Every retained example has a named-object certificate, and all tables distinguish exact/lower-bound/sample status.
- Every novelty claim has source-specific theorem comparison; X/priority limits remain disclosed.
- Old-paper corrections remain a separate unimplemented HOLD unless separately authorized; no claim is presented as new merely because it repairs an old defect.

Outcome: mathematical/novelty GO for a specific new-paper scope, NARROW, or STOP. Integrity PASS alone never satisfies this gate.

Phase9 — new manuscript, not a corrected MS03

Entry: Phase8 audit accepted and **explicit drafting approval**.
Work: write a new manuscript structured around the surviving core theorem; cite old and external work, include truthful limitations, source-linked proofs and reproducible examples. Do not reuse the old abstract's broad chain as the main contribution.

Measurable exit: statement-to-proof-to-example-to-source traceability; independent text/math consistency check; author approval of title, contribution claims, source/PDF hashes, declarations and data statement. If assumptions or theorems change, return to Phase8. No manuscript is drafted in the present task.

Phase10 — journal and publication decision

Entry: accepted new manuscript and current author/journal facts.
Work: verify journal scope, current quartile in a named database/year/category if Q1 is required, submission policies, related-work disclosure, data/licence/AI declarations and actual old-paper editorial status.

Measurable exit: source-dated journal/metric/policy records; all authors approve; no concurrent/redundant-submission conflict; exact files/source hashes identified. **Submission, email, deposit, release, commit or push each needs explicit authority.** Preparation files are not evidence of a submission or acceptance.

4. Conditional Q1-journal positioning

Candidate	Verified scope evidence	Fit if the proposed theorem succeeds	Main risk / exit condition
Designs, Codes and Cryptography — recommended first scope target	Official aims/scope, read as recorded in NJ01	Substantial algebraic coding theory, structural generator/hull classification with clear mathematical content	Routine Morita translation, weak novelty or predominantly uncompetitive tables would not justify a strong positioning claim
Finite Fields and Their Applications — conditional alternative	Official guide/aims, inspected NJ02 excerpts	Finite-field Frobenius/factor structure plays a **nontrivial** role in a theorem or meaningful algorithm	If the result is entirely general local algebra with field parameters incidental, scope fit weakens; overlap with NL01/NL03 especially important

Quartile status: neither journal is certified Q1 in this task. A future record must identify JCR/SJR/CiteScore or the author-required system, metric year, subject category, exact quartile and primary evidence. Do not infer a quartile from impact metrics or scope. No acceptance prediction, journal contact or submission is made.

If the theorem is correct but too routine for the aspiration, the honest choice is to deepen/re-scope or stop, not claim a Q1 guarantee. A lower journal target does not excuse invalid mathematics or hidden overlap.

5. Approval template for the next possible action

After reviewing 09, the author could explicitly select P2-S and authorize **Phase1 only: primary-literature, source-authority and scope review; no new proofs, scientific code, experiments, manuscript edits or Git writes**. This is an example of bounded authority, **not a recorded approval**. Any alternative scope must state its own boundaries.

Final status: BLUEPRINT DEVELOPMENT ONLY. Wait for explicit author approval before beginning Phase1.

09 — Author decision register for the new research blueprint

SOURCE / 09_AUTHOR_DECISION_REGISTER.md

All18 new decisions: PENDING — NOT APPROVED. No selection is entered on the author's behalf.

Master · Comparison · Phase plan

These **NAD** entries do not replace or resolve old AD01–AD20 in the consolidated audit. Arena recommendations are advisory. Scientific validity requirements cannot be waived by an author choice. A plan acknowledgement, preferred title or selected option is not automatically authority to execute another phase.

NAD01 — Primary research direction

Decision required: select a problem for further gated investigation, not a completed theorem.

Alternatives and consequences:

- **A: P2-S structural principal-input hull programme.** Gives a bounded algebra-first starting point; must pass external-gap and nonroutine-content checks.
- **B: P1 non-reflexive physical mechanism.** Greater direct quantum emphasis; requires differentiation from existing R1/EA/q8 work and hull-decomposition predecessors.
- **C: P3 skew/Galois compatibility.** Potentially deep but starts with quotient/side/form well-definedness and heavily occupied ingredients.
- **D: defer/stop.** No research begins; revise the comparison or await stronger literature evidence.

Arena recommendation: A, provisionally.
Mathematical justification: O-3.1b is explicit internally; a generator-profile criterion could differ from the known Morita threshold and nonprincipal-input witnesses. That is not an external novelty certificate.
State: PENDING — NOT APPROVED.

NAD02 — Exact prior source and editorial facts

Decision required: confirm MS03 as the actual current/submitted prior source, or identify a later exact source and actual publication/submission status.

Alternatives and consequences:

- **A: confirm MS03 with exact source identity and factual editorial status.** Enables a precise old/new overlap boundary; staging alone remains insufficient.
- **B: provide a later source/PDF lineage.** Requires re-inventory and renewed overlap/applicability assessment before dependent claims.
- **C: status/source not yet known.** Keep uncertainty explicit; block claims about submission, priority and definitive overlap completeness.

Arena recommendation: provide factual confirmation/identity rather than infer it from preparation directories.
Mathematical justification: a theorem may already be in a later private/submitted version; incorrect baseline selection risks duplicate claims or reliance on superseded statements. No editorial contact is authorized by this choice.
State: PENDING — NOT APPROVED.

NAD03 — Matrix size

Decision required: fix primary rank r.

Alternatives and consequences:

- **A: r=2 first.** Isolates a different principal-input structural problem with manageable representations.
- **B: r=3 first.** Requires μ(hull)>3 for nonprincipality, index9t and a larger budget; parameter substitution alone is insufficient.
- **C: all r.** Stronger scope but substantial risk of merely restating the known ceil(μ/r) threshold or making an unmanageable classification promise.

Arena recommendation: A; general r only as a justified later extension.
Mathematical justification: μ_S(I_W)=ceil(μ_B(W)/r) is already background. New content must concern the hull profile, not the packing threshold.
State: PENDING — NOT APPROVED.

NAD04 — Initial local/repeated-root class

Decision required: choose a bounded first theorem class.

Alternatives and consequences:

- **A: Core A, t=2,n=2,char2, q=2^m.** Clear single-factor starting point; may prove too small/routine.
- **B: a specified two-generated class at t=2,e=p^a>2.** Potentially richer structure but requires a precise class and transport contract before work.
- **C: arbitrary t,e and all embedded modules.** High representation complexity and weak feasibility; not recommended as a first phase target.

Arena recommendation: A, with a later author gate for B if needed; do not promise C.
Mathematical justification: dual-pairing constraints on principal inputs, rather than full wild classification, are the proposed source of tractability and differentiation.
State: PENDING — NOT APPROVED.

NAD05 — Twist scope

Decision required: decide which s-regimes the algebraic programme must cover.

Alternatives and consequences:

- **A: all0≤s<m in the chosen class.** Includes non-reflexive behavior; more intricate semilinear hypotheses.
- **B: reflexive twists only, m|2s.** Simpler symmetry but may lose the most differentiated Galois dependence.
- **C: s=0 first.** Easiest bilinear control; title/contribution must not pretend to establish general Galois behavior.

Arena recommendation: A as the question scope, with explicit restrictions if proofs demand them; no unproved all-s theorem.
Mathematical justification: DD=+σ_(2s) is inherited; its failure to be identity need not obstruct module hull analysis, but it forbids reflexive proof shortcuts.
State: PENDING — NOT APPROVED.

NAD06 — Acceptable mathematical outcome and stop rule

Decision required: accept preservation, growth, restriction or no-go as possible outcomes.

Alternatives and consequences:

- **A: outcome-neutral structural programme.** A meaningful preservation theorem or classified growth mechanism may qualify; routine/known outcomes trigger re-scope or stop.
- **B: investigate only a specified growth subquestion, without requiring a positive answer.** Narrower objective; a negative answer may end the project.
- **C: defer until a more differentiated question is available.** Avoids spending effort on an internally open but externally routine issue.

Arena recommendation: A.
Mathematical justification: no experiment or author preference can force μ(hull)>2 to occur. An isolated witness or a one-line known corollary does not meet the proposed paper-depth gate.
State: PENDING — NOT APPROVED.

NAD07 — Quantum component

Decision required: decide whether quantum applications are excluded, optional or separately pursued as central.

Alternatives and consequences:

- **A: optional after a valid structural/physical bridge.** Algebraic result stands independently; QG1–QG6 must all pass.
- **B: algebra-only.** Removes physical overhead and quantum claims; still needs substantial coding/algebra novelty.
- **C: quantum-led programme.** Requires a revised P1/NT09-centred plan and independent physical novelty review; cannot proceed merely by inserting h_s into EA formulas.

Arena recommendation: A, or B if no bridge survives.
Mathematical justification: μ_B(H_s), transported h_s, ordinary h_E and the trace-commutator radical are different invariants; standard EA/CSS routes are prior work.
State: PENDING — NOT APPROVED.

NAD08 — Separation from old-paper corrections

Decision required: preserve the independent old correction HOLD and define any later correction request separately.

Alternatives and consequences:

- **A: leave old work frozen; use only valid audited inputs with precise hypotheses.** New research remains distinct; old errors are not silently fixed.
- **B: pause the new programme and request a separate bounded correction task.** Needs its own source/claim/budget approval and does not count as the new paper.
- **C: request clarification of a disputed audit finding before either track.** A source-backed disagreement is reviewed; no theorem defect is waived.

Arena recommendation: A for this blueprint.
Mathematical justification: valid inherited theorems and defective printed proofs must be separated; a repaired old result is not new research.
State: PENDING — NOT APPROVED.

NAD09 — Literature access and novelty evidence

Decision required: choose how to obtain the primary theorem texts needed for Phase1.

Alternatives and consequences:

- **A: obtain accessible primary versions and author-provided legitimate copies where necessary.** Enables exact theorem comparison; source/version conflicts remain recorded.
- **B: remain with abstract/preview evidence.** Positive reported scope can be recorded, but dependent gap/priority claims stay X and may block GO.
- **C: defer the dependent question until access is resolved.** No invented citations or proof claims; slower but transparent.

Arena recommendation: A, with C for unresolved load-bearing sources.
Mathematical justification: NL03/NL10/NL11 may already imply the proposed result; absence from abstracts cannot establish a theorem-level gap. No cloud access or external editorial contact is authorized by this entry.
State: PENDING — NOT APPROVED.

NAD10 — Computational authorization and budget

Decision required: approve, reduce, enlarge or decline a future pilot's exact cases and resource cap.

Alternatives and consequences:

- **A: proposed bounded pilot: ≤3 families, ≤2,000 new inputs/family, total2 CPU-hours/4GiB/250MiB.** Limited falsification, no automatic exhaustive-family claim.
- **B: theory/source review only, no computation.** May avoid unnecessary enumeration; examples/certificates remain deferred.
- **C: a different explicit case list and cap.** Requires preflight combinatorial/memory estimates and a new approval; no unbounded default.

Arena recommendation: A only after Phases1–2 and separate implementation approval, otherwise B.
Mathematical justification: generator presentations and subspace families grow exponentially. More samples cannot prove a parameter-uniform conjecture.
State: PENDING — NOT APPROVED.

NAD11 — Independent review standard

Decision required: choose proof/computation reviewers and acceptance process.

Alternatives and consequences:

- **A: independent mathematical review plus separate arithmetic/certificate implementation.** Stronger error detection; requires reviewer effort before paper GO.
- **B: self-review and source-code agreement only.** Lower assurance; cannot be described as independent proof validation and may leave Phase8 blocked.
- **C: defer publication until external review is available.** Preserves research outputs without claiming submission readiness.

Arena recommendation: A, with C if necessary.
Mathematical justification: a shared wrong field library, Morita representation or proof dependency can make all tests agree. Universal claims require proofs, not finite agreement.
State: PENDING — NOT APPROVED.

NAD12 — Journal aspiration and ranking evidence

Decision required: choose a provisional scope target and the ranking system/year/category required for “Q1”.

Alternatives and consequences:

- **A: DCC as first scope target, contingent on surviving theorem.** Strong algebraic-coding fit; quartile still needs later primary verification.
- **B: FFA if finite-field/Frobenius/factor structure is nontrivial.** Appropriate only if the theorem fits that specific scope; close overlap with FFA predecessors must be addressed.
- **C: defer journal selection until Phase8.** Avoids optimizing for an unverified quartile or an as-yet-unknown theorem.

Arena recommendation: A provisionally, with final choice deferred and no Q1 certification now.
Mathematical justification: a structural explanatory theorem matters more than a journal label; neither scope nor impact metrics establish quartile or acceptance.
State: PENDING — NOT APPROVED.

NAD13 — Data, licensing and dissemination policy

Decision required: choose later storage/licence/disclosure policy for new research outputs without altering old artifacts.

Alternatives and consequences:

- **A: immutable small certificate bundles and manifests; larger data in an explicitly approved repository/storage, with author-approved licence.** Supports reproducibility without bloating Git.
- **B: private research storage initially.** Reuse/access limitations must be truthful; no public-data claim before release.
- **C: defer policy/release.** No automatic deposit, licence or public availability is asserted.

Arena recommendation: A at a later approved release gate; B/C meanwhile.
Mathematical justification: exact object/field/coordinate bytes are essential to verification; a public URL without a licence is not a reuse permission. This decision cannot retroactively change historical data statements.
State: PENDING — NOT APPROVED.

NAD14 — Field and coordinate convention

Decision required: fix future arithmetic encodings and any migration from inherited objects.

Alternatives and consequences:

- **A: retain actual inherited GF4/GF8/GF9 encodings for reused matrices and publish explicit new coordinate contracts.** Easiest exact-byte comparison.
- **B: use different explicit field models with checked isomorphisms.** Adds translation/certificate obligations; distances require correct coordinate interpretation.
- **C: defer implementation until conventions are fixed.** No ambiguous integer matrices are generated.

Arena recommendation: A for reuse; B only where scientifically justified.
Mathematical justification: isomorphic fields do not give identical integer labels; the documented old alternative encodings already caused mismatches. Morita/image/local vector lengths are also not interchangeable.
State: PENDING — NOT APPROVED.

NAD15 — Own-work overlap, authorship and truthful declarations

Decision required: confirm how the exact old work is cited/delimited and who is accountable for the new paper's claims/declarations.

Alternatives and consequences:

- **A: disclose MS03 and its status accurately, delimit inherited versus new results, confirm authorship/contributions/AI/data facts before drafting/publication.** Enables ethical related-work positioning.
- **B: wait for definitive old editorial/source facts before public positioning.** No submission-ready or priority statement while uncertain.
- **C: revise the proposed new-paper scope after overlap review.** May remove claims or end the project rather than conceal duplication.

Arena recommendation: A with B where facts are missing.
Mathematical justification: internal unpublished results still defeat a claim that the new project first establishes them; preparation folders do not prove submission or acceptance. No declarations are invented or copied as current facts.
State: PENDING — NOT APPROVED.

NAD16 — Conjecture posture

Decision required: choose whether PC1 is a formal proposed conjecture or only an open question.

Alternatives and consequences:

- **A: question-only until the exact-gap/applicability review.** No conjectural direction is assumed; preservation/growth remain symmetric possibilities.
- **B: retain PC1 explicitly as an exploratory falsification target.** Requires the full q=2^m,t=2,n=2,r=2,all-s scope and the limited q4 evidence label; it remains unproved.
- **C: defer conjecture formulation until after an approved pilot.** Additional finite evidence still would not prove a theorem.

Arena recommendation: A now.
Mathematical justification: the inherited1,221-module census covers a single field, not all m or all repeated-root multiplicities. A preferred outcome cannot be used as a proof assumption.
State: PENDING — NOT APPROVED.

NAD17 — Preservation and Git/publication actions

Decision required: retain current no-write/no-publication boundary or later request specific bounded actions.

Alternatives and consequences:

- **A: preserve all originals and seals; no commits, pushes, release, email or submission.** Current default; planning files are saved without Git writes.
- **B: later explicitly authorize committing/pushing an identified new-output set on the current session branch only.** Requires exact scope; does not authorize source corrections or a release.
- **C: later request a separately approved release/deposit/editorial action after scientific gates.** Needs current facts and exact files; preparation is not execution.

Arena recommendation: A now.
Mathematical justification: provenance must remain reproducible and historical errors visible;

integrity is not scientific or editorial authority. No branch switch/creation/rename is proposed.
State: PENDING — NOT APPROVED.

NAD18 — First-phase authorization

Decision required: explicitly authorize a bounded next phase, request blueprint revisions, or stop.

Alternatives and consequences:

- **A: after confirming the selected direction/scope, approve Phase1 only: primary-literature/source-authority/scope review, no new proofs, scientific implementation/experiments, manuscript edits or Git writes.** Produces a gap/scope decision; Phases2–10 remain unapproved.
- **B: request revisions to this blueprint.** Planning continues within that request only; no research execution.
- **C: do not proceed.** All new research remains paused; old correction HOLD unchanged.

Arena recommendation: review NAD01–NAD16 and, if satisfied, choose a bounded A; no such choice is recorded now.
Mathematical justification: the exact external gap and final theorem class are not certified. A literature-first gate prevents spending proof/implementation effort on routine or already known results.
State: PENDING — NOT APPROVED.

Approval record — currently empty

- Selected direction and initial mathematical scope: **NOT RECORDED**.
- Author source/submission confirmation: **NOT RECORDED**.
- Approved future phase/cases/budgets/files: **NONE**.
- Permission to prove theorems, implement/run science, edit a manuscript, commit/push or submit: **NONE**.

BLUEPRINT DEVELOPMENT ONLY. Wait for explicit author approval before beginning Phase1.

Appendix A — All 43 audit constraints

SOURCE / evidence/AUDIT_CONSTRAINT_CROSSWALK.csv

Every row retains the following source dispositions:
Implementation: NOT AUTHORIZED / NOT IMPLEMENTED
Closure: OPEN / AUTHOR CHOICE AND REVIEW REQUIRED

Severity counts: **8 CRITICAL · 29 MAJOR · 6 MINOR**. These are issue/evidence classifications, not counts of false theorems. CP/CR and RF are the original linked audit identifiers. Future-gate numbers are planning dependencies, not permissions. Exact evidence locators and all CSV columns remain in the attached source.

ID / severity	Original audit family / links	Constraint for new research	Future gates
MC01 MAJOR	**Right action and polynomial-module closure** CP01 / CR01 Classification: INCORRECT	Use the first-slot theta/right-action convention; verify shift/polynomial closure separately.	2
MC02 MAJOR	**Nondegeneracy probe versus perfect-duality argument** CP02 / CR02 Classification: INCORRECT	Do not infer perfect same-module duality from an invalid probe or nondegeneracy alone.	2
MC03 MAJOR	**Plain-product failure lemma and incompatible example data** CP03 / CR03 Classification: INCORRECT	Import no incoherent M2(F2)/uR/plain-product failure example or Fq proof for a generic ring.	2
MC04 MAJOR	**Slot/mirror convention and double-twist orientation** CP04 / CR04 Classification: INCORRECT	Keep positive DD in the primary convention; translate inverse/left/other-slot conventions explicitly.	1, 2
MC05 MAJOR	**Scalar/ring orthogonal equality and its printed proof** CP05 / CR05 Classification: INCORRECT	Ring/scalar orthogonal equality requires module closure and ideal-catching; no Hom-surjectivity shortcut.	2
MC06 MAJOR	**Unused A-valued candidate and inverse-parameter reconciliation** CP06 / CR06 Classification: INCOMPLETE	Do not add the unestablished A-valued pairing candidate as if equivalent to the selected dual.	2
MC07 MAJOR	**Base matrix ring versus cyclic ambient ring** CP07 / CR07 Classification: INCORRECT	Separate base matrix-ring ideals from modules over the repeated-root cyclic coefficient ring.	2
MC08 MAJOR	**Principality status, annihilator sides and cyclic/principal distinction** CP08 / CR08 Classification: INCORRECT	Treat cyclic and principal as distinct; the size-two global principality boundary is inherited, not open/new.	1, 2
MC09 MAJOR	**Chain-ring generation is not freeness** CP09 / CR09 Classification: INCORRECT	Use chain-ring generator bounds, not false freeness of every submodule.	2

ID / severity	Original audit family / links	Constraint for new research	Future gates
MC10 CRITICAL	**Nonprincipal-hull theorem's failed direct-sum family** CP10 / CR10 Classification: INCORRECT	The failed generic socle/direct-sum family has mu2; do not reuse its proof or call a repair new research.	2, 4, 8
MC11 MAJOR	**Phase5 hull row-representation bug** CP11 / CR11 Classification: INCORRECT	Separate full field-image rows from Morita vectors; require actual hull mu certificates.	2, 3, 6
MC12 MINOR	**Double-dual, reflexivity and generalization hypotheses** CP12 / CR12 Classification: INCOMPLETE	Distinguish global reflexivity from codewise invariance and do not extend arbitrary automorphisms verbatim.	2, 4
MC13 MAJOR	**Bilinear versus Hermitian congruence scope** CP13 / CR13 Classification: INCORRECT RF11	Keep bilinear, involutive Hermitian and general Galois congruences distinct.	2, 7
MC14 MAJOR	**Alternatingness is characteristic- and form-dependent** CP14 / CR14 Classification: INCORRECT	Do not infer alternatingness from zero diagonal without characteristic/form hypotheses.	2, 7
MC15 MAJOR	**GF9 transformation tested for the correct form** CP15 / CR15 Classification: INCORRECT RF06;RF11	Verify the actual transformed form and metric; GF9 bilinear evidence is not Hermitian evidence.	2, 3, 7
MC16 MAJOR	**Ambient obstruction is not a ban on every CSS code** CP16 / CR16 Classification: INCORRECT	An ambient congruence obstruction does not ban every individual CSS inclusion.	7
MC17 MINOR	**Scalar codomain and trace-level notation** CP17 / CR17 Classification: INCORRECT	Keep scalar Fq output, matrix-valued form and absolute Fp trace typed separately.	2, 7
MC18 MINOR	**Tautological Gray reflexivity diagnostic** CP18 / CR18 Classification: INCORRECT	Use the real divisibility diagnostic; a tautological header does not certify or refute later tests.	3, 6
MC19 MAJOR	**EXH/SMP and finite suite quantifiers** CP19 / CR19 Classification: INCORRECT RF06	State literal EXH/SMP/cartesian-pair scopes; assertion counts do not prove universal results.	3, 6, 8
MC20 MAJOR	**Generator field encoding** CP20 / CR20 Classification: INCORRECT RF08	Preserve actual GF8/GF9 integer labels or certify an explicit field isomorphism; no silent migration.	2, 3, 6

ID / severity	Original audit family / links	Constraint for new research	Future gates
MC21 MAJOR	**Historical census versus independent recertification scope** CP21 / CR21 Classification: COMPUTATIONALLY SUPPORTED ONLY RF03;RF04;RF05;RF07;RF08;RF16	Distinguish family enumeration, distance selection, finite certification and source execution.	3, 6, 8
MC22 MAJOR	**Odd-characteristic Weyl phase sign** CP22 / CR22 Classification: INCORRECT	Any quantum extension must use a verified odd-characteristic Weyl phase convention.	7
MC23 CRITICAL	**Stabilizer lemma hypotheses and nonbinary phases** CP23 / CR23 Classification: INCORRECT	Require trace isotropy and a phase-correct scalar-free stabilizer lift; arbitrary additive labels do not suffice.	7
MC24 CRITICAL	**EA Bob completion and Alice-only logical errors** CP24 / CR24 Classification: INCORRECT	EA requires valid Bob completion and Alice-only logical-error reasoning, not unextended noncommuting labels.	7
MC25 CRITICAL	**Route-specific label groups and orthogonals** CP25 / CR25 Classification: INCORRECT	Use the actual route label group and trace/Fq orthogonal; DC and R1 cannot share an unqualified formula.	7
MC26 MAJOR	**Euclidean hull versus transported Galois hull in EA** CP26 / CR26 Classification: INCORRECT	No substitution of transported Galois hull or module mu for ordinary h_E/physical radical.	7
MC27 CRITICAL	**DC relative quotient and coset-distance scope** CP27 / CR27 Classification: INCORRECT	DC uses its own relative logical-error set and dimension; ordinary EA/SO quotient is not universal.	7
MC28 MAJOR	**Coset-leader metric, projection proof and Singleton claim** CP28 / CR28 Classification: INCORRECT	Coset-leader metric is not representative Hamming weight; a corrected old bound is not a new theorem.	1, 7, 8
MC29 CRITICAL	**All-route bookkeeping identities** CP29 / CR29 Classification: INCORRECT RF02;RF03	Keep EA/SO/DC/R1 bookkeeping separate;742 DC literal failures are not742 false code distances.	3, 7, 8
MC30 MINOR	**Certification-method historical overclaim already fixed in MS03** CP30 / CR30 Classification: INCORRECT RF08	Retain the staged correction of all-three-methods wording; independently record each completed method and bound.	6, 7
MC31 MAJOR	**Extra wild/headline/R1 rows beyond the 28 certificates** CP31 / CR31 Classification: COMPUTATIONALLY SUPPORTED ONLY RF03;RF05;RF08;RF09;RF16	Every new named matrix needs its own certificate; old28 do not cover every row; an intersection is a subobject.	3, 6, 7

ID / severity	Original audit family / links	Constraint for new research	Future gates
MC32 CRITICAL	**Physical-alphabet benchmark reconstruction** CP32 / CR32 Classification: INCORRECT RF01	Match physical alphabet/resource/header, not a legacy construction-alphabet query parameter.	7, 8, 10
MC33 MAJOR	**Availability claims about q8 and EA tables** CP33 / CR33 Classification: INCORRECT	A missing guessed endpoint is not global table absence; q8 stabilizer and some EA tables exist.	7, 8
MC34 MAJOR	**Alahmadi DOI points to an unrelated article** CP34 / CR34 Classification: INCORRECT RF12	Use the correct Alahmadi .023 DOI; identity checking does not prove applicability.	1
MC35 MAJOR	**Other bibliographic identities, dates and missing DOI metadata** CP35 / CR35 Classification: INCORRECT RF10;RF11;RF12;RF13;RF14	Keep source/version metadata, author-order and unresolved technical cells explicit; do not invent DOI/status.	1, 8, 10
MC36 MAJOR	**Nearest-neighbour, priority and new-bound claims** CP36 / CR36 Classification: UNVERIFIABLE RF10;RF11;RF14;RF15	Compare exact predecessor theorems including A8; no firstness, new-bound or Q1 certificate from bounded searches.	1, 8, 10
MC37 MAJOR	**P1 'first q8 EA' proposal overlaps existing records** CP37 / CR37 Classification: INCORRECT RF15	P1 cannot be first-q8 EA; four supplied records already exist; require a new mechanism.	1, 7
MC38 MAJOR	**P2 size-three threshold/index/complexity claims** CP38 / CR38 Classification: INCORRECT RF15	P2 needs structural content; rank3 nonprincipal threshold is mu>3, index9t, base size p36 in the cited example.	1, 2, 5
MC39 MAJOR	**P3 Ore quotient and compatibility are not established** CP39 / CR39 Classification: INCOMPLETE RF15	P3 needs valid two-sided Ore quotient and sided twist/duality compatibility; combining ingredients is not novelty.	1, 2
MC40 CRITICAL	**Authoritative staging baseline, submission status and author-only gates** CP40 / CR40 Classification: UNVERIFIABLE RF15	Confirm actual source/submission facts; MS03 is staging authority only; retain old approval and editorial gates.	1, 8, 9, 10
MC41 MINOR	**86/88/87 file-count basis** CP41 / CR41 Classification: INCORRECT	State file-count bases and epochs; package/record counts are not interchangeable.	0, 6, 10

ID / severity	Original audit family / links	Constraint for new research	Future gates
MC42 MAJOR	**Qualified reference closure, unavailable binaries and build provenance** CP42 / CR42 Classification: INCORRECT	Partial reference closure and absent valid compiled PDFs do not prove complete recovery/submission identity.	1, 9, 10
MC43 MINOR	**Execution attribution and immutable audit history** CP43 / CR43 Classification: INCORRECT RF05;RF07	Keep inherited execution attributions/date discrepancies and seals immutable; no gratuitous campaign reruns.	0, 3, 6, 8

Appendix B — Source and preservation notes

This PDF is a presentation copy generated from the current local checkout. It does not commit, push, publish, reseal or change the source files. The current branch and unchanged checkout HEAD are provenance identifiers, not evidence that the untracked blueprint has been published to GitHub.

Branch: arena/7d88376d-matrix-ring
Unchanged HEAD: 168ec3e0b860cbe086ccb7a92a1a1aac5f8a277b
Sealed archive: NEW_RESEARCH_BLUEPRINT.zip
Archive SHA-256:
23dcc025177ddbf917abbd00c0d381a2e70d15
af56eeb53b11c31fa098e6b0b6

The ten numbered Markdown documents belong to the previously delivered 109-member ZIP. AUTHOR_FINAL_SCOPE_DECISION.md is a later supplementary document outside that seal. This PDF is also outside the seal. Neither addition updates the old manifest or implies author approval.

Embedded source attachments

The eleven main source Markdown files, the original package README and the full 43-row constraint CSV are attached byte-for-byte, together with a source-digest manifest. A PDF reader that supports attachments can extract them. Other historical context/evidence remains in the original ZIP and is not silently imported or re-endorsed.

- 10_MASTER_NEW_RESEARCH_BLUEPRINT.md
- AUTHOR_FINAL_SCOPE_DECISION.md
- 01_CANDIDATE_DIRECTION_COMPARISON.md
- 02_LITERATURE_AND_NOVELTY_POSITIONING.md
- 03_SELECTED_RESEARCH_PROBLEM.md
- 04_MATHEMATICAL_FRAMEWORK.md
- 05_THEOREM_ROADMAP.md
- 06_COMPUTATIONAL_METHODOLOGY.md
- 07_QUANTUM_CONSTRUCTION_STRATEGY.md
- 08_PHASE_WISE_EXECUTION_PLAN.md
- 09_AUTHOR_DECISION_REGISTER.md
- README.md
- evidence/AUDIT_CONSTRAINT_CROSSWALK.csv
- PDF_SOURCE_DIGESTS.json (generated presentation manifest)

Final status: BLUEPRINT / AUTHOR REVIEW ONLY. No Phase 1 execution. All author choices are pending. Original manuscripts, scripts, datasets and completed packages stay unchanged. Wait for the author’s explicit decision.